The Belgian Chamber Debates Dataset: Speeches, Roll-Call Votes and Member Biographies, 1831–2025

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Abstract

In this Data Descriptor, we introduce a dataset on the plenary speeches of the Belgian Chamber of Representatives and the background characteristics of the members who speak in it. The database includes all plenary speeches from the first sitting of the Chamber, in 1831, to the end of 2025, linked to the members and ministers who delivered them, together with the roll-call votes cast in the same sittings and a table of the bills and other legislative files that came before the Chamber, with their subject keywords, to which the roll calls are linked. These data have been used in previous studies to analyze how plural voting and proportional representation changed what the Chamber talked about and how far apart its parties spoke. The long time period, from census suffrage to the federal state, makes it possible to study how historical changes are reflected in parliamentary speeches.

Background & Summary

Parliamentary speech data can be used to study various political phenomena, including partisanship and political polarization[1–3], the effects of media and transparency on politicians' behavior[4], and political representation[5]. Unlike voting data, which are constrained by party discipline, parliamentary speeches can be used to better infer politicians' policy positions[6]. This is important in Belgium, where the parties control the parliamentary careers of their members and party discipline in the Chamber has traditionally been high[7,8]. Although party discipline may also affect parliamentary speeches, the parties' control of speeches is likely to be weaker than their control of votes, and previous literature has indeed found party discipline to be weaker in speeches than in votes, at least in Switzerland[9].

In this Data Descriptor, we describe a dataset covering all the plenary speeches of the Belgian Chamber of Representatives, from its first sitting in 1831 to the end of 2025. The data described here have been used, for 1880–1912, in two papers by the author and a co-author, which compare the Chamber with the Senate, less exposed to the reforms, in a difference-in-exposure design. In the first, we study how plural voting, introduced in 1893, and proportional representation, adopted in 1899, changed what the Chamber talked about[10]. The two reforms had asymmetric effects. After the first elections under plural voting, in 1894, the Chamber raised the share of its words devoted to labor and social policy by about 4.5 percentage points relative to the Senate. The effect was national—it came from outside the mining districts—and visible also on days when labor was not on the agenda, and it appears to come from members already in office, mostly Catholics, who adapted their speech. Proportional representation, first applied in 1900, did not widen the gap but moved it geographically: in the mining districts it changed who sat in the Chamber, mainly among the Liberals, rather than how sitting members spoke. In the second paper, we measure how distinguishable the parties are from their speech, with the estimator of Gentzkow, Shapiro and Taddy[1,11]. The elections of 1894 increased the partisan distance in the Chamber relative to the Senate, although with two chambers the evidence rests on a permutation test and remains suggestive. The reform of 1900 renewed the Chamber on a

large scale—more than a quarter of the words spoken came from newly elected members—but produced no detectable additional polarization. Read together, the first reform moved the Chamber as a whole, towards more labor in its speeches and more distance between its parties, while the second changed mainly who sat in it. The long time frame of the data is what makes such questions answerable: the reforms whose effects we want to understand are rare, and Belgium went through most of them within the period covered by the data (see Institutional setting).

Similar types of datasets have been used in other countries, such as Norway[2,12] and Finland[13], and have been built for Canada[14], Australia[15] and, for recent decades, a range of European parliaments[16,17]. Other projects have digitized the Belgian proceedings. Page facsimiles can be consulted online, the *plenum.be* project of the University of Antwerp has made the text of the plenary proceedings searchable[18], and the ParlaMint corpora cover recent legislatures[17]. The advantage of our data is that they are ready for analysis. They come in an accessible and compact format for analyses that use speeches from many different years, whereas searches targeted at specific speeches or sittings are better served by *plenum.be*. Every speech is linked to the characteristics of its speaker on the day it was delivered and to the roll-call votes of the same sittings; the roll calls are linked to the bills and other legislative files they decided, which carry the subject keywords recorded in the Chamber's legislative databases; and the text is kept as printed, so that users can apply their own preprocessing.

The Methods section describes the original data sources used to construct the dataset, the processing steps and the institutional setting. The Data Records section describes our datasets on parliamentary speeches, on the characteristics of the speakers and on roll-call votes and legislation, including the variables available. The Technical Validation section compares the data with well-known features of Belgian parliamentary history, and the Usage Notes outline possibilities for the use of the data. The Supplementary Information documents every step of the construction in detail and reports further validation.

Methods

Data sources and preprocessing methods

We use openly available digitized records of the plenary sittings of the Chamber as our main source. Data for 1831–2018 are available as text from *plenum.be* (`https://plenum.be`), which serves the text of the *Annales parlementaires*, and of the *Moniteur belge* before them, obtained by optical character recognition (OCR), together with page facsimiles[18]. From the 50th legislature, elected in June 1999, we take the plenary sittings from the PDF reports of the Chamber of Representatives (`https://www.dekamer.be`) instead: from October 2001 the reports print each intervention next to its summary in the other language, and the text of *plenum.be* runs the two together. From 2019 the roll calls come from the HTML version of the same reports and, for the 56th legislature, also from *mijnkamer.be*. Some files contain two sittings held on the same day, and the early volumes also contain sittings of the Senate: the 87 files of 1831–1848 that contain only a sitting of the Senate are left out. The process for transforming the OCR output into structured data files involves multiple steps tailored to different time periods.

First, the text of each sitting is stored as a single text file named after its date, and files that are not sittings—index volumes and tables of contents without the debate—are listed and left out. Next, each line is tagged using regular expressions such that it is then easier to separate the speeches from all other text: running heads, page numbers and footers; the heading of the sitting and the presiding officer; the table of contents; agenda headings; roll calls and the formulas of the clerks; and speaker labels and collective interjections. Tags for speakers are created from the speaker labels, which consist of a title, a surname, optionally a function, and a dash, as in *M. Tibbaut, rapporteur.* —. For example, speaker labels are tagged with expressions of the following form (simplified):

```
(?P<title>M\.|MM\.|Mme|De heer|Mevrouw)\s?
(?P<name>[A-ZÉÈ][\w'-]+(?:\s+(?:de|van|der)?\s*[A-ZÉÈ][\w'-]+)*)
(?P<function>,\s*(?:rapporteur|ministre[^--]{0,90}))?
\s*[.,]?\s*--
```

The expressions tolerate the forms that the OCR produces for the title—*AI.*, *MA.* or *33.* for *M.*—and follow the formats of each period, from the labels closed by a colon in the *Moniteur* to the numbered labels of the modern reports. As the OCR output contains some errors, the name printed in each label is matched to the closest name among the members of the Chamber in office on the date of the sitting, and otherwise among the ministers in office, allowing for misread characters, particles and compound surnames. The names that remain unlinked are listed by frequency, and we have resolved the most frequent by hand; these decisions are released with the data. The registers of members and ministers are reconstructed from the Dutch-language Wikipedia, which lists the members of every legislature and the composition of every government. Turns of the chair are attributed to the presiding officer named in the heading of the sitting and in the notices of changes of chair; from 2019, when a report does not give that name, to the President of the Chamber in office that day.

In the third step, the text of each sitting is split into blocks at these tags. The process is the following: (1) screen and date the files; (2) tag each line, in a fixed order of precedence, as page furniture, heading of the sitting or presiding officer, table of contents, agenda heading, roll call or formula of the clerks, or speaker label, and tag labels in the middle of a line when several turns are printed on the same line; (3) split the sitting into blocks at the tags and give each block a type— `SPEECH`, a turn by an identified speaker; `CROWD`, a collective interjection; `PROC`, text of the clerks; `DOC`, a document read into the record; `VOTE`, a roll call; and, from 1999, `SUMMARY`, the summary of an intervention in the other language—while headings, tables of contents and page furniture are stored in a separate table; (4) read the lists of each roll call into one row per member and vote; (5) link each speaker and voter to the registers; (6) extract biographical variables for the members; and (7) link each roll call to the legislative file it decided.

The final step assigns each block its language. A block is in Dutch when its label uses a Dutch title (*De heer*, *Mevrouw*) or when its text contains at least two words from a list of frequent Dutch function words and more of them than of French ones, and in French otherwise; where the text interleaves an intervention with its summary in the other language, each line is assigned to the one in its language. Unlike datasets that distribute lowercased and stemmed text[13], we keep the text of every block exactly as printed—OCR errors included, with only the soft hyphens at line breaks removed—together with the pages on which it appears, and leave preprocessing to users. Biographical variables—birth, death, noble title, education, occupations and other political offices—are extracted from the Dutch and French Wikipedia articles on each member by a large language model, Claude Haiku 4.5, whose output is constrained to a fixed schema and which must support every value with a verbatim quotation, then searched for in the article; for 8 members the values come from the open-weight Gemma 3 model[19], run locally with the same instructions.

The legislative files come from the Chamber's own legislative databases[20]. For 1831–1988, the *Genesis* collection holds a sheet for every bill and other item of business before the Chamber and the Senate, with its title, its authors, the parliamentary documents that belong to it, the dates of its votes in the two houses and subject keywords. From 1988, the FLWB database records for every file of the Chamber its title, its status, the dates of its deposit and of its votes in the two houses, keywords and EuroVoc descriptors and, for laws, the date of the law and of its publication in the *Moniteur belge*. We download the sheets and pages of every legislature from the site, merge the two sources into one table with one row per file, and link each roll call to the file it decided. From 1999 the report prints the number of the file next to most numbered votes. Before 1999 no number is printed, but every bill had to be voted on as a whole by roll call (see Institutional setting), and the databases give the date of the Chamber's vote on each file: a roll call is linked to a file that the Chamber voted on the same day and, when it voted several, to the one whose title best matches the agenda heading and the text around the vote; a roll call with no such file is linked, if its text matches a title closely, to a file voted in the same or one of the next three years.

The share of block boundaries recovered by the automated rules is 84.7 %, with a precision of 93.6 %, in 37 of the 39 excerpts of a gold standard annotated by hand, one from a sitting drawn at random in each five-year period, and 83.4 % of their text carries the right type and speaker. The rules were revised after a first evaluation, and the numbers may therefore be read as the accuracy of the final version (Supplementary Information S4). Each step is implemented in a separate script, described in detail in Supplementary Information S1.

Institutional setting

The Belgian parliamentary system is party-centered. Since 1900 the members of the Chamber have been elected by list proportional representation in multi-member districts, and the parties control the selection of candidates, the order of the lists and, to a large extent, the behavior of their members in parliament[7].

Belgium is a particularly informative case for long-run analyses. The Constitution of 1831 created a bicameral parliament elected under a restrictive census suffrage. A constitutional revision in 1893 introduced universal male suffrage tempered by plural voting, together with compulsory voting; in 1899 Belgium became the first country to elect its national parliament by list proportional representation, first applied in 1900; plural voting gave way to equal male suffrage in 1919; and women voted in national elections from 1949 (Supplementary Table S1)[7,21,22]. Over the twentieth century, the conflict between Dutch- and French-speakers turned the unitary state into a federation through a sequence of state reforms between 1970 and 1993[7]. The Chamber kept a verbatim record of its sittings throughout[23], so that the parliamentary side of each of these transitions is documented in a single and remarkably stable source.

The Chamber of Representatives is the lower house of the Belgian Parliament. Until the state reform of 1993 it shared legislative power on an equal footing with the Senate, whose proceedings are not part of this dataset. Votes may be taken by sitting and standing, but the Constitution of 1831 required a roll call on the whole of every bill (Art. 39)[24], as the coordinated Constitution of 1994 still does (Art. 55)[25], so that every bill on which the Chamber voted as a whole should have left a roll call in its record. Its members were elected in multi-member districts (*arrondissements*). Until 1919 the Chamber was renewed by halves every two years, so that its composition changed at every partial election. The Chamber grew with the population, reaching 212 members in 1949, before being reduced to 150 members from 1995 onward[26]. The Chamber did not sit during most of the German occupations. After an extraordinary sitting on 4 August 1914, it did not meet again until 22 November 1918. During the Second World War, parliamentary meetings in Brussels stopped after 10 May 1940 and resumed on 19 September 1944[27,28]. Ministers may address the Chamber whether or not they belong to it, a right established by Article 88 of the 1831 Constitution[24] Art. 88. Some of the ministers who spoke there were therefore senators or held no seat in parliament. We consequently link speakers to a register of ministers as well as to the register of members.

For more than a century and a half the verbatim report was drafted from shorthand note. In October 2000 it was renamed *Compte rendu intégral* (*Integraal Verslag*) and shorthand gave way to the transcription of audio recordings[23]. From 1999 each intervention is numbered and accompanied by its summary in the other national language. Supplementary Table S1 lists the institutional changes that matter most for users of the data.

Data Records

Dataset on parliamentary speeches

Table 1 describes the dataset on parliamentary speeches, `speeches_linked.csv`. The data cover the years 1831–2025, except for most of the years of the German occupations, 1914–1918 and 1940–1944, when the Chamber did not sit. Each row is a block of the debate, in the order of the printed report: `sitting` and `date` identify the sitting, `seq` the position of the block within it, and `type` whether the block is a speech, an interjection, text of the clerks, a document, a roll call or a summary. The column `label` reproduces the speaker label as printed, OCR errors included; `speaker_as_printed` gives the name read from it, and `speaker` the name normalized against the members who took part in the roll calls of the surrounding sittings. For turns of the chair, `speaker` holds the presiding officer. The columns `page_start` and `page_end` give the pages of the printed volume on which the block appears, so that any passage can be located in the facsimiles; `section` gives the agenda heading under which it appears, which allows speeches to be grouped into debates; and `lang` gives its language, French or Dutch.

The column `person_key` links speakers to the dataset on speaker characteristics, and `link` records

Table 1. Description of the dataset on parliamentary speeches (`speeches_linked.csv`; one row per block of the debate).

Column	Description
`sitting`	Identifier of the sitting (file stem, `DD_MM_YYYY`)
`date`	Date of the sitting (YYYY-MM-DD)
`doc`	Document code of the report, from 1999 (e.g. `CRIV 54 PLEN 051`; `COM` for committee meetings)
`seq`	Position of the block within the sitting
`type`	`SPEECH`, `CROWD`, `PROC`, `DOC`, `VOTE` or `SUMMARY`
`label`	Speaker label as printed, OCR errors included
`speaker_as_printed`	Name as printed in the label
`speaker`	Name normalized against the roster; for turns of the chair, the presiding officer; the group (e.g. *gauche*) for `CROWD`
`role`	Function stated in the label (e.g. *président*, *rapporteur*, *ministre de l'intérieur*)
`party`	Party printed in the label (from the 1990s)
`lang`	Language of the block (`fr`, `nl`)
`section`	Agenda heading under which the block appears
`page_start, page_end`	First and last page of the block in the printed volume
`flags`	Space-separated warnings on doubtful decisions: e.g. `not_in_roster` (name not among the members of the period), `function_only` (label with a function and no name), `ocr_title`, `colon` or `no_dash` (label in a non-standard form), `resumed` (turn resumed without a label), `by_role` or `by_context` (speaker resolved from the function or the context)
`text`	Text of the block as in the source, soft hyphens removed
`person_key`	Identifier of the member or minister (joins the dataset on speaker characteristics)
`leg_label`	Legislature in office on the date of the sitting
`link`	How the speaker was linked (see text)
`min_portfolio`	Portfolios held on that day, for ministers
`m_*`	Mandate on the date of the sitting: party, district, roles, replacement or departure
`pol_*`	Register entry of the member (`politicians.csv`)
`bio_*`	Biographical variables (Supplementary Table S6)

how the link was made: by name, exactly (`exact`) or allowing for OCR errors (`fuzzy`); by hand (`fix`); to a minister by name (`minister`) or by portfolio (`minister_role`); or, from 2019, to the President of the Chamber in office that day (`chair_president`). Rows that remain unlinked are marked `not_found`, `ambiguous`, `gender_mismatch`, `no_legislature` or `chair_unnamed`. Linked rows carry the member's mandate on the date of the sitting (`m_*`), the member's register entry (`pol_*`) and the biographical variables (`bio_*`). The column `text` holds the text of the block as printed; because the linked file repeats it for every block, the file can also be written without the text, which can then be joined back from `speeches.csv` on `sitting` and `seq`.

Dataset on speaker characteristics

Table 2 describes the dataset on speaker characteristics, `members_year.csv`, which has one row per member, legislature and calendar year of the legislature (46,262 rows, 3,138 persons). It offers metadata that can be used to study, for instance, the descriptive representation of different groups in the Chamber. The column `person_key` links it to the dataset on speeches, and `year` and `leg_label` give the year and the legislature; because legislatures follow one another in election years, a member re-elected in such a year appears once for each legislature. The columns `party` and `district` record the party and electoral district as listed for the member in that legislature, and `family` groups the party labels into party families for descriptive purposes. The column `mandate_type` tells members listed in the tables of the legislature from those named only in the notes of another member, such as members who took their seat as replacements.

Table 2. Description of the dataset on speaker characteristics (`members_year.csv`; one row per member, legislature and year).

Column	Description
`person_key`	Identifier of the member (joins `speeches_linked.csv`)
`leg_label`	Legislature (e.g. 1892–1894)
`year`	Calendar year of the legislature
`party, district`	Party and electoral district, as listed for the legislature
`family`	Party family (Catholic / Christian democrat, Liberal, Socialist, Flemish nationalist, Francophone regionalist, Green, Communist / radical left, Other), for descriptive use
`mandate_type`	`table` (listed member) or `notes` (named only in the notes of another member)
`female`	1 for women, 0 for men (Wikidata)
`birth_year, age`	Year of birth; age in the year
`highest_education`	Highest level of education (Supplementary Table S6)
`university`	1 if the highest level is a university or postgraduate degree
`noble_title`	Noble title, as written
`minister`	1 if the member held a ministerial office in the year
`n_prev_legislatures`	Legislatures in which the member sat before this one
`first_year`	First year of the member's first legislature
`experience`	Years since `first_year`

The column `female` records the gender of the member, from Wikidata; the columns `birth_year`, `age`, `highest_education`, `university` and `noble_title` come from the biographical variables; and `minister` indicates whether the member held a ministerial office at any time in that year. Finally, as in the Finnish data, we have calculated variables describing the experience of the members: `n_prev_legislatures` counts the legislatures in which the member sat before the current one, `first_year` gives the first year of the first of them, and `experience` the number of years since then. The file is built from the registers of members and ministers and from the biographical variables, which are released in full (Supplementary Tables S5 and S6).

Unlike the Finnish data[13], the dataset does not include election results; they can be joined through the party and the district of each member. It would also be possible to link these data to the party codes of ParlGov or PartyFacts. Doing so, however, involves choices that are better left to researchers: party labels change over time, cartels list several parties under one name, and the national parties split along linguistic lines from the late 1960s, so that mapping them onto codes requires decisions about which parties to treat as the same entity.

Dataset on roll-call votes and legislation

Three files describe how the members voted and on what. The file `votes_linked.csv` has one row per member and roll call: `sitting` and `rollcall` identify the roll call, `vote` gives the member's vote (`oui`, `non`, `abstention` or `présent`), `rollcall_status` records whether the names listed agree with the counts announced in the report, and `person_key` and the columns `m_*`, `pol_*` and `bio_*` link the voter to the dataset on speaker characteristics, as for the speeches (Supplementary Table S4).

The file `laws.csv` (Table 3) has one row per legislative file (*dossier*)—a government bill, a member's bill, a budget, a naturalization or another item of business—from 1831 to 2025: 47,924 files, 21,504 of which were voted by the Chamber. Each file is identified by `law_id`, its code in the Chamber's databases (as in `09K0093-1857-1858` or `54K3441`), and carries its title in French and, where available, in Dutch, its type, its authors (for the Genesis files), the dates of its votes in the Chamber and in the Senate, subject keywords (`keywords`) or, from 1988, EuroVoc descriptors (`eurovoc`), and, where the source gives them, the date of the law and of its publication in the *Moniteur belge*. The column `is_law` marks the 19,464 files that became law, and `law_basis` the evidence: the date of the law or, before 1995, when a bill needed the vote of both houses to become law, a date of vote in each house.

The file `rollcall_laws.csv` links the two. It has one row per roll call before 1999 and one per item of a numbered vote from 1999, with the file decided (`law_id`), the object of the vote (`kind`: the whole of a bill, an article, an amendment, a motion and so on), whether it is a vote on the whole of the file on the day of the Chamber's vote or, from 1999, a numbered vote on a bill as a whole (`final`), and how the link was made (`method`; see Methods and Supplementary Information S1). Joined to `votes_linked.csv` on `sitting` and `rollcall`, it gives the members' votes on each file wherever they were recorded by name (from 1999, `rollcall` is empty for the numbered votes without a list of names); joined to `laws.csv` on `law_id`, the subject of each roll call. Votes on motions and on the order of business usually decide no file and are then left without one.

Other files

The other files follow the same conventions. From 1999, `vote_items.csv` gives the file, the object, the outcome and the counts of every numbered vote as printed in the report, from which `rollcall_laws.csv` takes its links, and `law_documents.csv` lists the parliamentary documents of the Genesis files. The apparatus of the printed record—headings, presiding officers, tables of contents and page furniture—is kept in `apparatus.csv` rather than discarded, and `sittings.csv` documents the date of every sitting. The registers of members and ministers, the biographical variables with their supporting quotations, and the diagnostics and manual decisions that document each step are released as well (Supplementary Information S2).

Technical Validation

We plot the number of speeches per year and the length of speeches over time in Fig. 1. The series respond to the events known to have shaped the work of the Chamber. The number of speeches (Panel a) rises from a few thousand a year in the 1830s and early 1840s, when the source covers the sittings only as far as the digitized issues of the *Moniteur* go, to between 5,000 and 10,000 a year from the late 1840s to the 1870s and 10,000 to 20,000 in the 1880s. It jumps after the first elections held under plural voting, in 1894, and stays between 15,000 and 40,000 a year until 1914. It falls to zero during the German occupations of 1914–1918 and 1940–1944, when the Chamber did not sit, and to its lowest post-war values in 1992–1995, when part of the plenary business went to a Central Commission whose proceedings are not included (Supplementary Information S4). The year with the most speeches is 1913, with 39,942 speeches. The length of speeches follows the practice of the record (Panel b). From the 1890s to the 1960s, when the report records every exchange on the floor, the median speech is only about 20 words long; it lengthens from the 1970s, and the numbered interventions of the modern reports, transcribed from audio recordings from 2000, have a median of 150 to 200 words. Overall, the median speech was 36 words long in 1831–1893 and 166 words in 1995–2025. The very long speeches of the 1830s and 1840s partly reflect turns that the rules fail to separate in the oldest volumes, where recall is lowest (Supplementary Table S12).

Regarding the dataset on speaker characteristics, Fig. 2 confirms what is already known. Women appear among the members from the legislature that began in 1929—the first woman was elected to the Chamber in 1929—but remain below about 10 % until the 1990s. Their share rises steeply from 1995 and reaches 41.2 % in the last legislature (Panel a). This increase coincides with the introduction and subsequent strengthening of gender quotas on electoral lists: the 1994 Smet–Tobback law limited candidates of the same sex to two thirds of a list, while the 2002 parity law required an approximately equal number of men and women[29,30].

The share of members with a university degree (Panel b) rises in the nineteenth century to about three quarters, falls at each extension of the suffrage—after the first elections under plural voting in 1894 and after the introduction of equal male suffrage in 1919, when the Belgian Workers' Party brought workers and union officials into the Chamber—and rises again from the late 1960s. It is 60.5 % in the first legislature for which biographies are complete and 78.5 % in the last. Panel c shows that most members have always been aged 40 or more when their legislature began: at least 57.3 % in every legislature, the lowest shares

Table 3. Description of the files on legislation: `laws.csv`, one row per legislative file, and `rollcall_laws.csv`, one row per roll call before 1999 and per item of a numbered vote from 1999. The individual votes (`votes_linked.csv`) are described in Supplementary Table S4.

Column	Description
laws.csv — one row per legislative file	
`law_id`	Code of the file in the Chamber's databases: legislature, `K` (Chamber) or `S` (Senate), number and, before 1954 and for some later files, session (e.g. `09K0093-1857-1858`, `54K3441`)
`source`	`genesis` (printed lists), `genesis + sheet` (with the full sheet), `genesis sheet` (full sheet only) or `flwb`
`legislature, chamber`	Legislature; house of the file (`K` or `S`)
`type, initiative`	Type, read from the title (`projet de loi`, `proposition de loi`, `budget`, `naturalisation`, ...); initiative, read from the first author or, without one, from the type (`government`, `member`, `committee`, ...)
`title, title_nl`	Title in French; in Dutch, where available
`authors, first_author`	Authors as printed in Genesis (surname and given name)
`status, date_depot`	Stage reached and date of deposit (FLWB and full sheets)
`vote_chambre, vote_senat`	Dates of the votes in the Chamber and in the Senate
`date_loi, moniteur_date, moniteur_no`	Date of the law; date and number of its publication in the *Moniteur belge*
`reached_senate, voted_both`	The file went to the Senate; it was voted by both houses
`is_law, law_basis`	1 if the file became law; evidence: `law date`, or `both chambers` (votes in both houses, before 1995)
`keywords, eurovoc`	Subject keywords (Genesis and some FLWB files); EuroVoc descriptors (FLWB)
`commission`	Committee that examined the file (FLWB)
`documents, n_documents`	Parliamentary documents of the file (for the Genesis files, one row each in `law_documents.csv`)
`url`	Page of the file on the Chamber's website
rollcall_laws.csv — one row per roll call; from 1999, per item of a numbered vote	
`sitting, date, doc`	As in `speeches_linked.csv`
`rollcall, vote_no, item`	Roll call within the sitting (joins `votes_linked.csv`); official number of the vote and item, from 1999
`kind, final`	Object of the vote (`whole`, `amendment`, `article`, `motion`, `urgency`; from 1999 also `bill`, `budget`, `resolution`, ...); 1 for a vote on the whole of the file on the day of the Chamber's vote and, from 1999, for every numbered vote on a bill as a whole
`yes, no, abstentions, outcome`	Counts—of the names listed before 1999, as announced from 1999—and outcome (from 1999)
`section, headings, vote_text`	Agenda heading or title of the vote, the last headings before it, and the text of the vote or its conclusion, as printed
`law_id`	File decided by the vote (joins `laws.csv`); empty when none was found
`law_title, law_type, law_vote_chambre`	Title, type and date of the Chamber's vote of that file
`method`	How the file was found: `document number` (from 1999), `vote date`, `vote date + title`, `title`; otherwise `ambiguous` or `none`
`score, n_candidates, candidates`	Share of the title's words found around the vote, weighted by their rarity among the files voted that day (unweighted for `title`; empty from 1999); number of files voted that day; files tied, for `ambiguous`

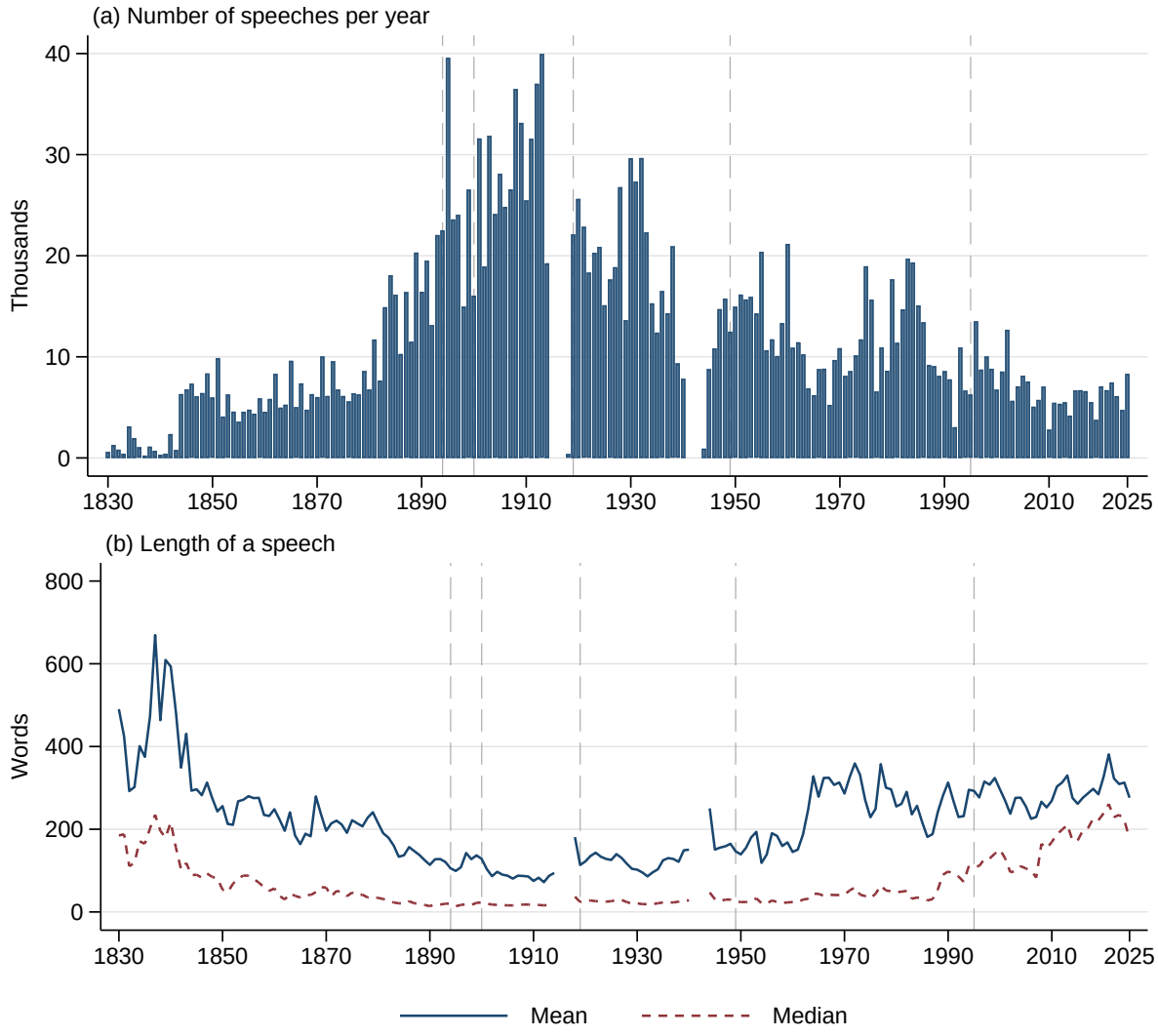


Fig. 1. Number of speeches and speech length. (a) Speeches per year: blocks of type `SPEECH` in plenary sittings. (b) Words per speech, mean and median by year, turns of the chair excluded. Dashed lines: the reforms of 1894, 1900, 1919, 1949 and 1995.

being those of the first legislatures, in the 1830s, and of the legislature elected in 1894 [31,32].

The roll calls and the legislative files can be checked against each other (Fig. 3). Because every bill had to be voted on as a whole by roll call, every file that the Chamber voted according to its own databases—almost all of them bills—should have a roll call on the day of the vote. The number of files voted (Panel a) falls to almost zero during the German occupations, as the number of speeches does. From 1999, when the report prints the number of the file next to most votes, 99.6 % of the files voted by the Chamber have a roll call on the day of the vote and 98.8 % are linked to one (Panel b). Before 1999 the two shares are 73.7 % and 43.7 %. They are highest in 1930–1979, when 96.7 % of the files voted have a roll call on the day of the vote, and lowest before 1930, when only 43.9 % do: the roll calls of the older reports are incompletely detected, and their counts should be read as lower bounds (Supplementary Information S4).

Beyond these comparisons with known history, we evaluated each step of the construction (Supplementary Information S4). Against the gold standard, the rules reach an F1 of 88.9 % on the boundaries between blocks, and attribute 83.4 % of the text to the right type and speaker (Supplementary Table S12). Overall, 95.6 % of the speeches and 95.9 % of the individual votes are linked to a member or a minister (Supplementary Table S13), and in a random sample of 200 linked speeches checked by hand, 97.5 % are linked to the right person. Supplementary Information S4 also reports the dating of the sittings, diagnostics computed on every sitting, the consistency of the roll calls, the links between roll calls and legislative files, by decade and by method, and the biographical variables.

Usage Notes

These data have been used in previous studies to analyze how plural voting and proportional representation changed the topics of parliamentary speech [10] and the partisan distance between the parties [11]. The long time period makes it possible to study how historical changes are reflected in parliamentary speeches: the data lend themselves, for instance, to measuring polarization from speech over almost two centuries [1,3], to comparing what members say with how they vote on the same bills, to studying the volume and subjects of legislation through the keywords of the legislative files, and to studying how the enfranchisement of new voters changed who spoke, about what and in which language. Practical notes for users—on the unit of analysis, the coverage and quality of each period, the biographical variables and the licences—are given in Supplementary Information S7.

Code availability

Stata code to produce the tables and figures in this article (the do-files `00_master.do` to `08_laws.do`), and the Python code that builds the dataset from the sources, are released with the dataset. The biographical variables were extracted with Claude Haiku 4.5 (model `claude-haiku-4-5-20251001`, Anthropic Message Batches API) and, for the members for whom it returned no valid answer, with Gemma 3 12B (Ollama tag `gemma3:12b`, model ID `f4031aab637d`).

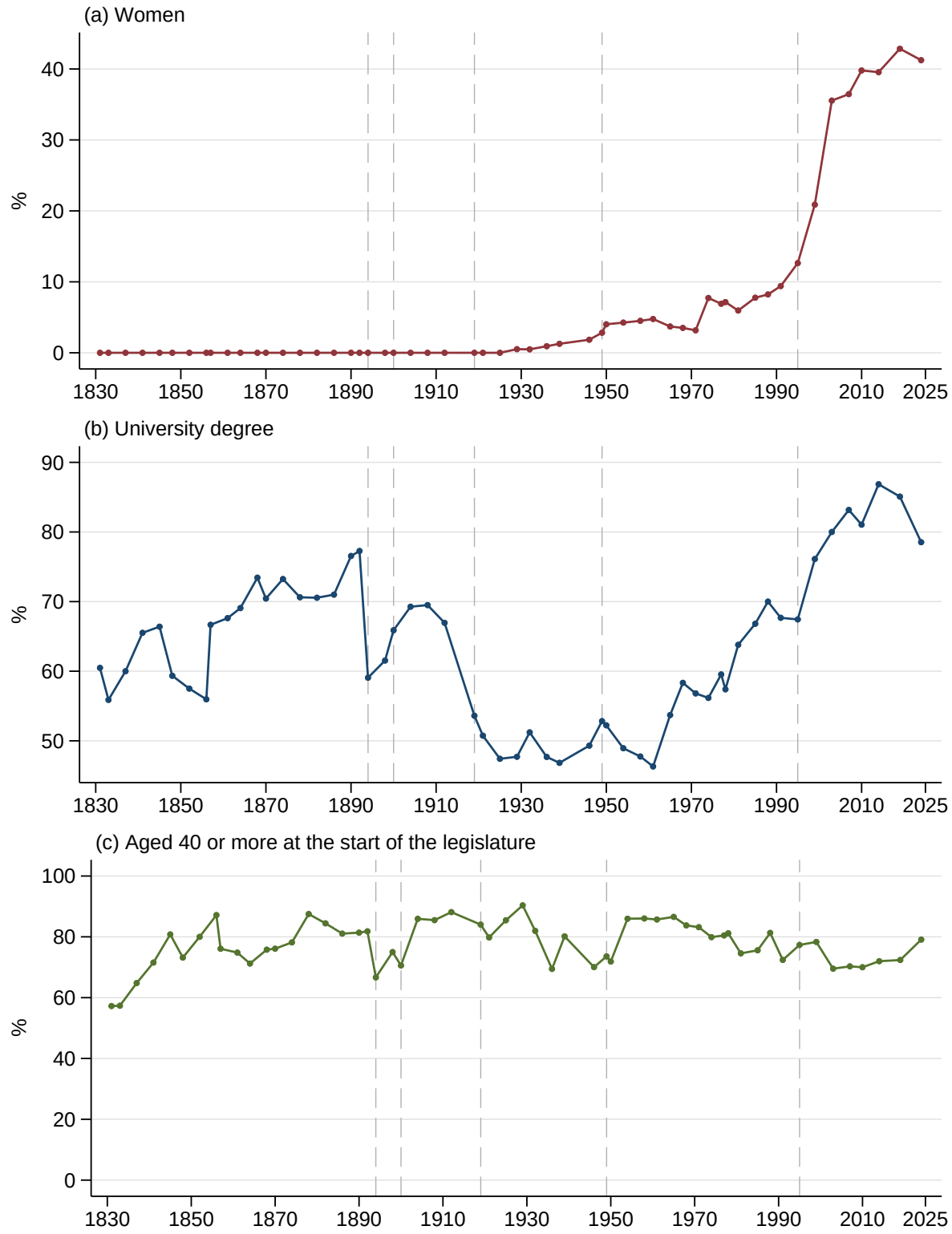


Fig. 2. Characteristics of the members, by legislature. (a) Share of women among the members. (b) Share with a university degree, among members with biographical variables. (c) Share aged 40 or more at the start of the legislature. Panels b and c show only legislatures in which at least 80 % of the members have the variable.

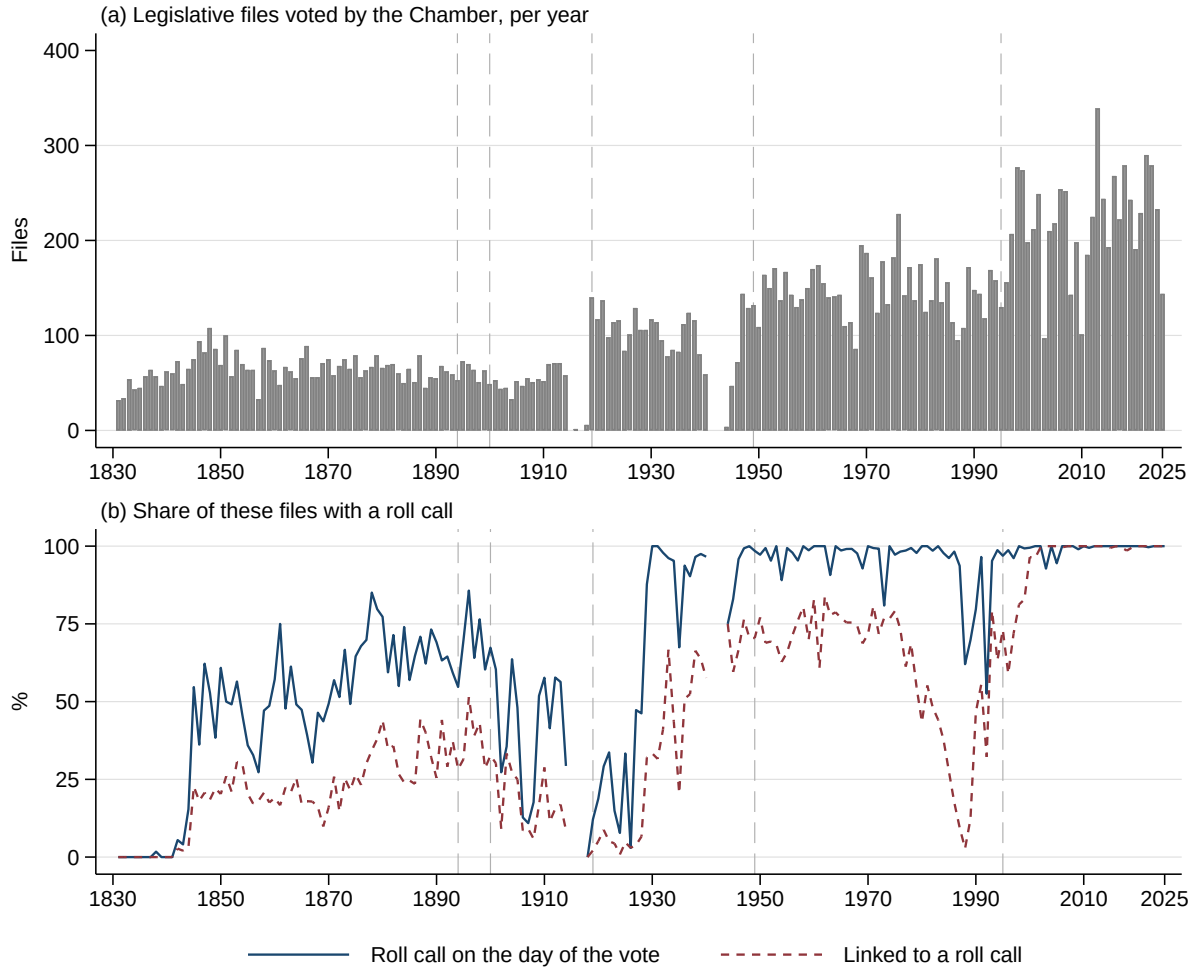


Fig. 3. Legislative files voted by the Chamber and roll calls. (a) Files whose sheet in the Chamber's legislative databases gives the date of the Chamber's vote, by year of the vote. (b) Share of these files for which the data contain a roll call on the day of the vote, and share linked to at least one roll call in `rollcall_laws.csv`. Dashed lines: the reforms of 1894, 1900, 1919, 1949 and 1995.

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Author contributions

Riccardo Graziani designed the dataset, wrote the code, built and validated the data, and wrote the manuscript.

Competing interests

The author declares no competing interests.

APPENDIX

Supplementary Information

This supplement documents every step of the construction of the data (S1), the files and variables not described in the main text (S2), summary statistics (S3), further validation (S4), stylized facts (S5), the modules in preparation (S6) and notes for users (S7). Table S1 lists the institutional changes that matter most for users of the data.

Table S1. Institutional changes that matter for users of the data.

Year	Change	Consequence for the data
1830–31	National Congress, the constituent assembly, meets from November 1830 and adopts the Constitution on 7 February 1831	Its sittings precede the first legislature and are not linked to the register
1831	Constitution establishes a bicameral parliament and census suffrage; deputies are elected by majority rule in multi-member districts, and the Chamber is renewed by halves every two years	Start of the Chamber; membership changes at each partial election
1893–94	Universal male suffrage with plural voting and compulsory voting, first applied in the 1894 elections	The Belgian Workers' Party enters the Chamber in 1894
1898	Equality Law gives the Dutch and French texts of legislation equal legal status	Language of each block recorded in `lang`
1899–1900	List proportional representation using the D'Hondt method, introduced in 1899 and first applied in the 1900 elections	Several parties can obtain representation within the same district
1919–21	Equal universal male suffrage first applied in the 1919 election and incorporated into the Constitution in 1921; the Chamber moves from partial to complete renewal	Legislatures of up to four years
1929	First woman elected to the Chamber	Female titles (*Mme*, *Mevrouw*) appear in speaker labels
1948–49	Women's suffrage in national elections, introduced in 1948 and first exercised in a legislative election in 1949	The electorate includes women from the 1949 legislature onward
1970–93	Four state reforms, in 1970, 1980, 1988–89, and 1993, transform Belgium from a unitary into a federal state	Language groups are introduced; the powers, composition, and procedures of the Chamber change
1993–95	Fourth state reform reduces the Chamber to 150 members, effective from the 1995 election	The Chamber has 150 members from the 1995 legislature onward
1999–2000	Digital recordings begin to replace shorthand transcription in 1999; in 2000 the *Annales* are renamed *Compte rendu intégral*	Numbered interventions and summaries in the other language

S1. Construction of the data

Four principles guided the construction of the data. First, the historical text is never rewritten: the segmenter decides only where blocks begin and what they are, and the text of each block is kept as printed. Second, nothing is silently discarded: text that is not debate is routed to the apparatus table, files that are not sittings are listed with the reason for their exclusion, and doubtful decisions are recorded as flags rather than resolved by fiat. Third, every record can be traced back to its source: blocks carry the pages of the printed volume on which they appear, sittings the date printed in their heading, member records the revision of the Wikipedia page from which they were read, and every model output the model version and a hash of the prompt that produced it. Fourth, every automated step is evaluated, against human annotation where possible and against internal consistency checks everywhere.

Fig. S1 summarizes the construction of the dataset. Two tracks, built from different sources, produce the debates and the registers of members and ministers, and a record-linkage step joins them; a third track reads the legislative files of the Chamber and links them to the roll calls. Each step is implemented in a separate script and can be rerun independently of the others.

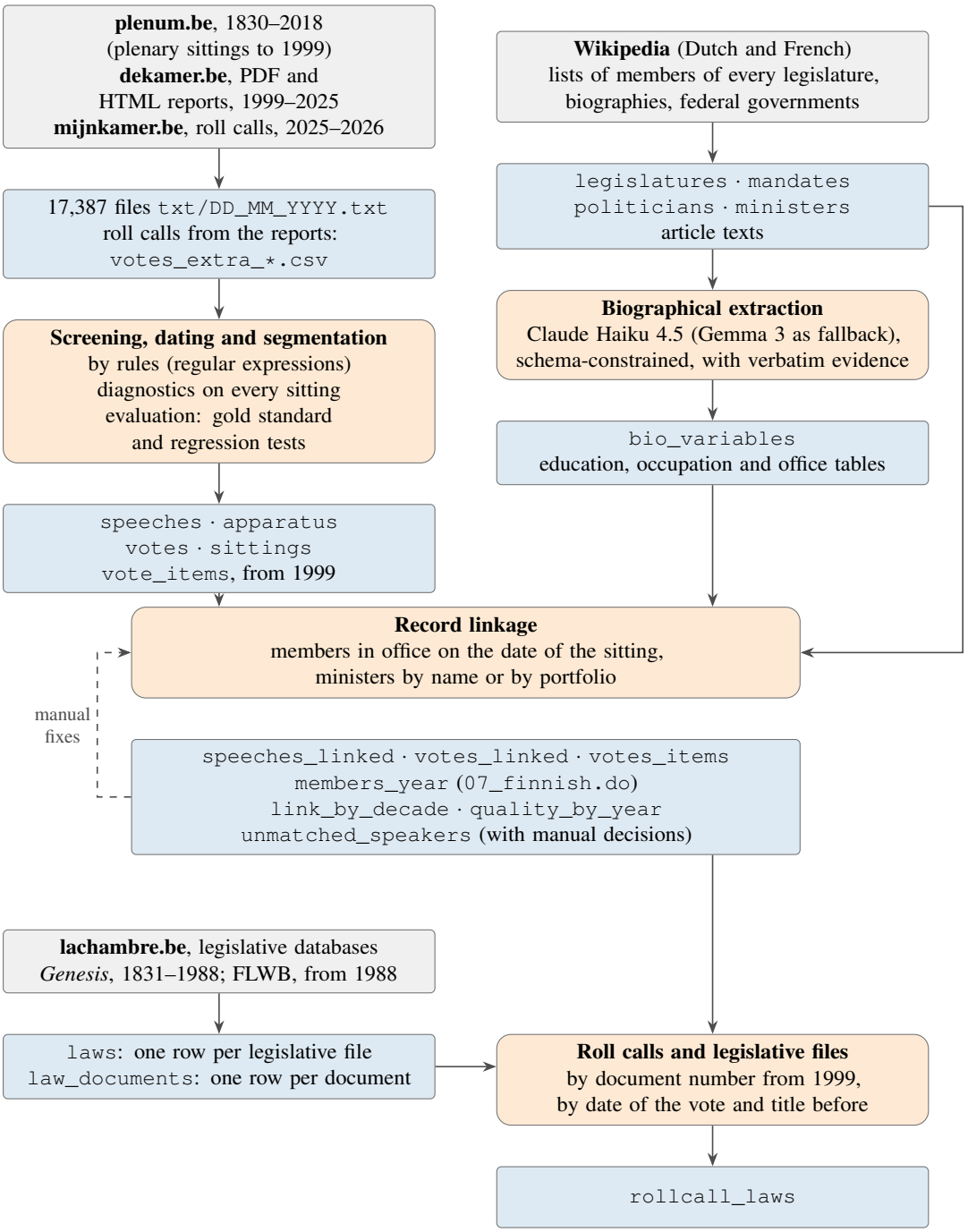


Fig. S1. Construction of the dataset. Grey: sources; blue: files released with the dataset (CSV unless stated otherwise); orange: processing steps. The debates (left) and the registers of members and ministers (right) are built independently and joined by date-aware record linkage; the legislative files (bottom) are read from the Chamber's legislative databases and joined to the roll calls by document number or by the date of the vote and the title. Names that cannot be linked automatically are resolved by hand in `unmatched_speakers.csv`, and these decisions are released with the data. The Senate module (1880–1914) goes through the same steps.

Element	Example as printed in the *Annales* (1910, abridged)	Stored in
Sitting heading	SÉANCE DU MERCREDI 2 FÉVRIER 1910.	`apparatus`, `sittings`
Presiding officer	PRÉSIDENCE DE M. COOREMAN, PRÉSIDENT.	`apparatus`
Table of contents	SOMMAIRE. — Communications. — [. . .], p. 421.	`apparatus`
Agenda heading	COMMUNICATIONS. FIXATION DE L'ORDRE DU JOUR.	`apparatus`, `section`
Procedural text	La séance est ouverte à 2 heures. — Pris pour information.	`PROC`
Speaker turn	N. Tibbaut, rapporteur, — [. . .] *(the title "M." misread by the OCR as "N.")*	`SPEECH`
Collective interjection	A GAUCHE : [. . .] DES VOIX A DROITE : [. . .]	`CROWD`
Document read into the record	Art. 1er. [. . .] *(article of a bill, amendment, table)*	`DOC`
Roll-call vote	— Il est procédé au vote par appel nominal. *n* membres y prennent part. [. . .] Ont répondu non : MM. [. . .] Ont répondu oui : MM. [. . .] Se sont abstenus : MM. [. . .]	`VOTE`, `votes`
Page furniture	424 CHAMBRE DES REPRÉSENTANTS. — ANNALES PARLEMENTAIRES. ANN. PARL. — CH. DES REPR. — SESSION ORDINAIRE DE 1909-1910.	`apparatus`

Fig. S2. Elements of a sitting and where they are stored. The examples follow the typography of the sitting of 2 February 1910; text is abridged, and the speaker label reproduces an actual OCR error. `PROC`, `SPEECH`, `CROWD`, `DOC` and `VOTE` are block types of `speeches.csv`; from 1999 a sixth type, `SUMMARY`, holds the summary of an intervention in the other national language. The other elements go to `apparatus.csv`, and each agenda heading is also copied into the `section` column of the blocks that follow it.

Source: the *Annales parlementaires*

The Chamber has produced a verbatim report of its plenary sittings since its first session in 1831. The report was printed in the official gazette, the *Moniteur belge*, until 1844, and thereafter as a separate series, the *Annales parlementaires* (in Dutch, *Parlementaire Handelingen*). For more than a century and a half it was drafted from shorthand notes; in October 2000 the *Annales* were renamed *Compte rendu intégral* (*Integraal Verslag*), and shorthand gave way to the transcription of audio recordings[23]. From 1986 the public meetings of the Chamber's committees received their own *Annales*[23]; until the 1991–92 session, however, the fascicles of these meetings were issued within the series of plenary *Annales*[33], which matters for the composition of the corpus (see Usage Notes). The digitized series also contains the sittings of the National Congress, the constituent assembly that met from November 1830 to July 1831 and adopted the Constitution.

The printed report follows a recognizable grammar (Fig. S2). Each sitting opens with a heading that gives the date and the presiding officer; from about 1876 it is followed by a table of contents (*sommaire*) that lists the items of business with page references. The debate itself is a sequence of turns, each introduced by a speaker label made of a title, a surname, optionally a function, and a dash (for example, *M. Tibbaut, rapporteur.* —). Turns alternate with agenda headings in capitals, procedural formulas written by the clerks, collective interjections from the benches, texts read into the record—bills, amendments, tables—and roll-call votes that list the members who voted yes, voted no or abstained. Pages are printed in two columns with running heads and footers, which the OCR places in the flow of the text, sometimes in the middle of a sentence. French was the dominant language of the debates in the nineteenth century; interventions in Dutch became more frequent over time. The format changed at several points, which the segmenter follows period by period (see below).

Retrieval of the text

1830–2018: plenum.be. We use the OCR text published by *plenum.be*, a research website maintained by the Political History research group of the University of Antwerp, which makes the proceedings of the Chamber searchable alongside page facsimiles[18]. We use this text as served and do not correct it. The site is served by a public application programming interface (API) that lists sittings by year, at most 200

per request, and returns for each sitting its full text and the text of its individual pages. Our client walks through the list year by year and downloads the full text of every sitting, falling back on the page-level text when the full text is unavailable. Requests that fail for transient reasons are retried; sittings that the server reports as missing are not retried but recorded in a summary report. Such gaps occur, for instance, in the 54th legislature (around 2015), whose plenary sittings we take from the Chamber's reports (see below). The text of each sitting is stored unmodified as a UTF-8 file named after the date of the sitting (`DD_MM_YYYY.txt`). A date may have more than one sitting, and for some dates the file held a committee meeting, the week's table of contents or a sitting of the Senate instead of the plenary sitting. Each file is therefore checked against the sittings that the site lists for its date. A sitting counts as present when at least half of its blocks of 150 words are found in the file, a block being found when at least a quarter of its sequences of four consecutive words are. Sittings that are not present are added in the order of the day, and a file that holds none of the plenary sittings of its date is replaced: 1,751 files were completed or replaced in this way, 288 of them for dates that had no file. Sittings of the Senate, annexes (*Documents*) and tables of contents that the site lists among the sittings are not added, nor, up to 1914, a sitting whose text is already in the file of a date at most three days away, since the site and the volumes do not always date a sitting alike. From November 1985 the files also contain the reports of public committee meetings, numbered C1, C2 and so on; from 1999 each report carries the Chamber's document code, which distinguishes plenary sittings (e.g. *CRIV 54 PLEN 051*) from committee meetings (*CRIV 54 COM 286*), and the segmenter stores this code with every block. Before 1999, when no code is printed, the reports of public committee meetings and of the Central Commission are recognized from their titles (S7).

1999–2025: the Chamber's verbatim reports. From October 2001 the report prints each intervention in a wide column, next to its summary in the other national language in a narrower one. The text of *plenum.be* runs the two columns of each page together, often in mid-sentence, so that an intervention is followed by its summary. For the plenary sittings we therefore use the PDF files of the *Compte rendu intégral* published by the Chamber on *dekamer.be*, one per sitting, from the first sitting of the 50th legislature, in July 1999. They replace the plenary sittings of the files of *plenum.be* for the same dates: 895 days of 1999–2018 are written from 1,222 reports, and 306 days of 2019–2025 from 417 reports. A day is written from the reports of all its plenary sittings, provided that every report carries the date of the day on its first page; the committee meetings of its file follow, in the text of *plenum.be*. Seven days of 2000–2005 keep the text of *plenum.be*, because the site does not list their sittings or dates them differently from the reports. Each day is written as one file in the same format as above, every sitting headed by its document code. On each page the gap between the two columns is the widest vertical band, between 35 and 80 % of the page width, crossed by the fewest words; we keep the lines that begin to the left of it, so that lines printed across the whole page—announcements of the president, titles of questions, votes and their lists of names—are kept whole, and leave out the lines that begin to its right (text extracted with PyMuPDF, falling back on pdfplumber). In the earlier reports every column holds the verbatim record—two columns, three in July 1999—and from October 2000 to July 2001 the summary follows on pages of its own: there we keep every column, in reading order, and leave out the pages of the summary.

Roll calls from 2019. From the 55th legislature (2019–2024), the names of the members who took part in each roll call are printed in an annex whose multi-column layout the text extraction does not preserve. We therefore take the lists from the HTML version of the same reports, which keeps the lists intact: we identify the 239 plenary reports of the 55th and 56th legislatures that contain roll calls, retrieve them at a rate of one every 1.5 seconds, and read the lists of names and the document numbers printed next to each vote. For the 56th legislature we also use *mijnkamer.be*, an independent civic-tech project that republishes roll-call data from the Belgian Chamber's official website, *dekamer.be*[34]. The project states that its data are synchronized directly from the Chamber's plenary reports and made available as open data free of charge; its public pages do not identify a separate institutional operator or specify a distinct reuse licence[34,35]. It publishes each roll call with the party of every member; we downloaded 1,005 roll calls and 135,599 individual votes cast in 50 plenary sittings between 18 June 2025 and 17 September 2026. The dataset ends

on 31 December 2025, and later sittings are left out. The votes from these sources are written to separate files (`votes_extra_criv.csv`, `votes_extra_mijnkamer.csv`) and replace, report by report, the votes read from the text; when a report is covered by both, the Chamber's own report takes precedence.

Screening and dating. Of the 17,387 files, 107 are not sittings and are left out: 80 are index volumes (such as the *Notice chronologique*) and 27 contain only a table of contents without the debate, all of them from the 1990s (see Usage Notes). Both kinds are detected automatically and listed, with the reason, in `skipped_files.csv`. The *Moniteur belge* of 1831–1848 printed the sittings of the Senate in the same volumes as those of the Chamber, and some files contain a sitting of the Senate. The two houses are told apart by their running heads (*SÉNAT, SÉANCE DU* and *CHAMBRE DES REPRÉSENTANTS*): the 87 files that contain only a sitting of the Senate are left out, and the 149 files that contain both houses are kept and flagged, their Senate part unlinked (`senate_files.csv`). The remaining 17,280 files are sittings, from 12 April 1830 to 18 December 2025. For each sitting, the date printed in its heading is read and compared with the date in the file name (Table S14). When the two differ because day and month are swapped in the file name—92 files, mostly from the nineteenth century—the printed date is used; in every other case the date of the file name is kept, because the heading read by the OCR may be corrupted or belong to a neighbouring sitting, and the discrepancy is recorded. `sittings.csv` reports, for every sitting, the date used, the date of the file name, the printed date and the outcome of the comparison.

Segmentation of the debates

Each sitting is segmented by a rule-based parser that uses only the Python standard library. We chose rules over a segmenter based on a large language model (LLM), which we implemented as a benchmark (see below), because they are deterministic, fast—the whole corpus is segmented in a few hours on a laptop—and fully inspectable: every decision can be traced to a single regular expression.

Cleaning. Soft hyphens left at line breaks (Unicode U+00AD, with any white space that follows them) are removed, so that words split across lines are rejoined. No other change is made to the text of the blocks: OCR errors are kept as they are, and case and accents are normalized only when names are compared.

Line-level parsing. The parser reads a sitting line by line and assigns each line to the first category that matches, in a fixed order of precedence: (1) page furniture—running heads with page numbers, footers and isolated numbers; (2) the heading of the sitting and the presiding officer, including changes of chair during the sitting; (3) the table of contents, from its heading to the formula that opens the sitting (*La séance est ouverte*); (4) agenda headings, short lines in capitals that end with a period and may run over two lines; (5) roll calls and the formulas of the clerks; and (6) speaker labels and collective interjections. A line that matches none of these continues the current block. Because several turns are often printed on the same line, a speaker label may also open a block in the middle of a line; labels that fall inside a quotation are ignored. The regular expressions tolerate the OCR errors that recur in the corpus: the running head *CHAMBRE DES REPRÉSENTANTS* is recognized when letters inside *CHAMBRE* are misread, and the heading *SOMMAIRE* when its first letters are, as in *sumMAIRE*. Page numbers read from the running heads are carried forward, so that every block records the first and the last page on which it is printed.

Speaker labels and names. A speaker label consists of a title, a name, an optional function or qualifier (e.g. *rapporteur*, *ministre des finances*) and a closing dash or colon. The titles are *M.* and *MM.* in French and *De heer* in Dutch; for women members, from 1929, *Mme*, *Mevrouw*, *Mevr.* and *Mej.*; and a function in capitals when the speaker is a minister named only by office (*M. le Ministre de la Justice*). The rules accept the corrupted forms that the OCR produces for the title *M.*, such as *AI.*, *RI.*, *SI.*, *NE.*, *ni.* and *III.*, which alone accounted for about 40,000 missed labels in 1840–1914 before they were added. We store the label as printed and, separately, the name, the function and, when the label prints it, the party. Names are

normalized against a roster built from the roll-call lists of all sittings within two years of the current sitting: a printed name is mapped to the most similar roster name, and the roster also validates labels whose title was lost in the OCR. When roster names are consolidated across sittings, OCR variants are clustered under the constraint that two names printed in the same roll call cannot refer to the same member, which keeps members with similar names, such as Hymans and Huysmans, apart. A name that cannot be matched is kept as printed and flagged `not_in_roster`.

The presiding officer. Turns of the chair are printed as *M. le président* or *De voorzitter*, without a name. The parser reads the name of the presiding officer from the heading of the sitting (*PRÉSIDENCE DE M. COOREMAN*, *VOORZITTER: De heer Langendries*) and from the notices of changes of chair during the sitting, and attributes the turns of the chair to that person, who can then be linked like any other speaker. The rules cover the forms used from 1832 to 2025: headings split over two lines, headings that give only the first name of the presiding officer, the name printed on the line after the heading, and OCR variants such as *Ie Président*. Each presiding officer found is recorded in the apparatus table. The name is also read from the opening formula of the modern reports (*La séance est ouverte à 14 h 19 et présidée par Mme Eliane Tillieux, présidente*); when a report of 2019–2025 names no presiding officer, the turns of the chair are attributed at the linkage stage (see Record linkage).

Block types. Every block of the debate receives one of six types: `SPEECH`, a turn by an identified speaker; `CROWD`, a collective interjection such as *A GAUCHE :* or *DES VOIX A DROITE :*, whose speaker is the group; `PROC`, text written by the clerks without a speaker (openings and closings, communications, excuses, formulas such as *Pris pour information*); `DOC`, a document read into or printed in the record, such as the articles of a bill, an amendment or a table, which the first version of the rules attributed to the chair who introduced it; `VOTE`, a roll call; and `SUMMARY`, from 1999, the summary of an intervention in the other national language, which carries the speaker of the intervention it summarizes. Each block also records the agenda heading under which it appears, which allows users to group speeches into debates, and, from 1999, the document code of the report. Headings, tables of contents, running heads and footers form the apparatus table.

Period-specific grammars. The rules were first developed on sittings from 1880–1914 and then extended, period by period, to the whole corpus, using targeted samples of sittings for each change of format and the diagnostics described in Technical Validation. In the *Moniteur belge* of 1830–1844, labels end with a colon rather than a dash (*M. JULLIEN :*). Before about 1876, the heading of a sitting does not name the day of the week and the table of contents is not closed by the opening formula. From 1929, women members are introduced by the titles listed above, and in the following decades long documents inserted in the record have to be kept apart from the interventions of the chair, which is the purpose of the `DOC` type. In the 1990s, labels end with a colon and give the party in parentheses. From 1999, interventions are numbered, as in *02.01 Name (Party):*, and each is accompanied by its summary in the other language; the parser recognizes the numbered labels and stores the party. The plenary sittings read from the verbatim column of the PDF reports contain only the interventions; in the text of *plenum.be*, which we keep for the committee meetings and for seven plenary days of 2000–2005, each summary follows its intervention under a label that repeats its number and speaker, and the parser assigns it to a `SUMMARY` block when the number is that of an earlier intervention and the text is in the other language. The language of each block (`lang`) is identified by the rule described in Methods: Dutch when the label uses a Dutch title or the text contains at least two frequent Dutch function words and more of them than of French ones, French otherwise; where the text interleaves interventions with their summaries, each line goes to the intervention or to the summary in its language.

Regression testing. Every revision of the rules was compared with the previous version on a stratified sample of 20 sittings per decade. For each decade, the comparison reports the number of speeches found by each version, the share of speech text that is left unchanged, the text moved to `DOC` and `SUMMARY` blocks,

and the sittings in which the new rules find fewer speeches than the old ones, which were inspected by hand; both versions receive the same roster, so that the comparison isolates the rules.

Benchmark segmentation with a language model

As an alternative to the rules, we implemented a segmenter based on an LLM. To exclude the possibility that a generative model alters the historical text, the model never reproduces it. Each sitting is divided into chunks of about one printed page, and every line is prefixed with its number. For each block in a chunk, the model returns the number of the line on which the block starts; an anchor, that is, the opening characters of the block copied exactly as printed, OCR errors included; the type of the block; and, for speeches, the speaker, the function, the language and a flag for uncertain decisions. Python code then checks every anchor against its line, cuts the text, removes page furniture and joins blocks that continue across page breaks. The prompt defines each block type and includes three worked examples taken from the sitting of 2 February 1910, which is therefore excluded from the evaluation sample. Calls are made at temperature 0, with output constrained to a JSON schema when the provider supports it, and the provenance of each call is logged (`llm_calls.jsonl`). At the default chunk size, segmenting the whole corpus would require 471,758 calls, at an estimated cost of about US$390 for a small hosted model with prompt caching. This benchmark has not yet been evaluated on the gold standard: the released data are segmented by the rules, which are evaluated in S4.

Roll-call votes

A roll call opens with a fixed formula (*Il est procédé au vote par appel nominal*; in the modern reports *Naamstemming* or *Vote nominatif*), followed by the announced counts (*n membres y prennent part*; *n répondent oui*; *n répondent non*) and by lists of names introduced by *Ont répondu oui :*, *Ont répondu non :* and *Se sont abstenus :*. In the modern reports the lists are printed in an annex at the end of the report, whose format changed several times: from 1995 to 1999, for instance, each list is introduced in both languages (*Ont répondu oui – Voor hebben gestemd : Les membres – De leden :*), and from 2019 the counts are printed on three separate lines. While a list is open, lines that look like speaker labels are kept in the list. When the report states that a roll call repeated the previous one (*Même appel nominal que le précédent*), the lists of the previous roll call are used. The parser splits each list into names, normalizes them against the roster and writes one row per member and roll call, with the vote (`oui`, `non`, `abstention` or `présent`). From 1999 roll calls carry an official number within the sitting, which is stored as `vote_no`. For every roll call, the announced counts are compared with the numbers of names found in the lists, and the outcome is recorded in the apparatus table and with every vote (`rollcall_status`): `ok` when the counts agree, `mismatch` when they do not and `no_counts` when no counts are found; roll calls whose lists could not be read (`no_list`) have no individual votes. Explanations of abstention that follow a roll call are treated as speeches. From 2019 the votes read from the text are replaced by those of the Chamber's HTML reports and of *mijnkamer.be* (see Retrieval).

Votes and legislative files

From 1999, the report prints the parliamentary document concerned by each numbered vote, identified by the number of the legislative file (*dossier*) and of the document within it, as in *(3441/10)*. The reference is printed either in the heading that announces the vote or, as in 2003–2007, in the conclusion that follows it (*Bijgevolg neemt de Kamer het wetsontwerp aan*); a heading belongs to the next vote and a conclusion to the previous one. We read these references from the text of the reports and write one row per numbered vote (`vote_items.csv`), with the dossier and the document, all the references cited, the object of the vote (an amendment, an article, a bill, a motion, a budget and so on), the outcome, the announced counts, the place where the reference was found and whether heading and conclusion agree. For the reports of 2019–2026, the references of 4,966 votes in 263 reports are taken from their HTML version (see Retrieval).

A second file, `votes_items.csv`, attaches these fields to the individual votes. Dossier numbers restart in every legislature, so that a dossier is identified by the legislature and its number (as in the Chamber's own notation, *DOC 54 3441*). Of the 13,629 numbered votes, 88.6 % are linked to a dossier, and so are 87.4 % of the individual votes cast since 1999. The dossiers themselves—their titles, subject keywords, stages of the procedure and, for laws, the dates of the law and of its publication—are described in `laws.csv` (see the next section).

Legislative files

The Chamber documents its legislative work in two databases on its website[20]. The *Genesis* collection covers 1831–1988 and holds 38,812 sheets, one for every bill and other item of business before the Chamber and the Senate: government bills (*projets de loi*), members' bills (*propositions de loi*), budgets, naturalizations, reports on petitions, addresses to the King and the tables of each session. A sheet gives the title of the file, its authors, the parliamentary documents that belong to it, the dates of its votes in the Chamber and in the Senate and free subject keywords (*mots-clés libres*). Its code gives the legislature, the house (`K` or `S`) and the number of the file, followed by the session until the files began to be numbered through the legislature in 1954, and for some later files (`09K0093-1857-1858`, `36K0352`). The site serves the sheets through a search form whose results are written by scripts in the browser, so we drive a visible Chrome window through it with Selenium: for each legislature we search all the sheets of both houses, save the printed list of the results, which gives titles, authors, documents and dates, and page through the results, 50 sheets at a time, for the keywords, which the printed list omits. The search returns at most 2,000 sheets, a limit reached for the 42nd and the 45th legislatures, whose remaining sheets are missing. The same lists, requested in Dutch, give the Dutch titles. For 1,903 files, we also download the full sheet, which adds the status of the file, the date of its deposit and, for laws, the date of the law and of its publication in the *Moniteur belge*.

From 1988 the files of the Chamber are in the FLWB database, one page per file. We open the pages legislature by legislature, file number after file number until the numbers run out, in browser windows, since the site shows the files only to a browser that runs its scripts, at most one page every three seconds in each window; the pages are saved as they are and parsed in a second step, so that the parser can be revised without downloading them again. A page gives the title, the status, the dates of deposit and of the votes in the two houses, the committee, the documents with their authors, keywords and EuroVoc descriptors and, for laws, the date of the law and of its publication in the *Moniteur belge*. Both downloads wait and try again when the site refuses pages, stop when it asks for a CAPTCHA, and resume where they stopped.

We merge the sources into `laws.csv`, one row per file: the sheets of the printed lists, completed by the full sheets where they exist, and the files of the FLWB. We leave out the sheets whose code marks them as an index of a session and those without documents; some lists and indexes with an ordinary code remain, with the type `table`. The type of each file is read from the beginning of its title, and its initiative from its first author—the government (*Gouvernement*), a member (a person's name), a committee (*Commission*) or the clerks (*Greffe*)—or, for the FLWB files and the full sheets without a printed list, from its type (`government` for a *projet de loi*, `member` for a *proposition de loi*). The parliamentary documents of every file of the printed lists are listed in `law_documents.csv`; for the FLWB files, the column `documents` gives the main documents. A file is marked as a law (`is_law`) when its sheet gives the date of the law or of its publication in the *Moniteur belge* or when, before 1995, both houses voted on it, as a bill then needed the vote of both to become law; petitions, addresses, accounts, tables and requests for naturalization are never marked on the second ground, and the rare bill that the second house rejected is marked too, since a date of vote does not say how the vote went. The table contains 47,924 files, 36,292 from Genesis and 11,632 from the FLWB; 21,504 carry the date of a vote of the Chamber, and 19,464 are marked as laws.

Each roll call is then linked to the file it decided (`rollcall_laws.csv`). From 1999, most numbered votes carry the number of their file (`vote_items.csv`, see above), which gives the link directly (`document number`). Before 1999 the link rests on the date. The sheets give the day on which

the Chamber voted each file, and on that day the file was voted on as a whole by roll call; the candidates for a roll call are therefore the files voted on the day of the sitting, and the text around the vote decides among them: the agenda heading under which the vote is printed, the end of the block before it and the beginning of the vote itself. Each candidate is scored by the share of the words of its title, in French or in Dutch, found in that text, each word weighted by $\log(1+n/n_w)$, where n is the number of candidates and n_w the number whose title contains the word, so that the words shared by all the treaties approved on the same day weigh little; only the part of a title before its first parenthesis counts, without the words common to all titles (*projet*, *loi*, *relatif* and the like). A single candidate is accepted when it scores at least 0.15, or whatever its score when the roll call is a vote on the whole (`vote date`); among several, the best is accepted when it scores at least 0.3 and at least 0.1 more than the next (`vote date + title`); when other files score within 0.1 of it, the one whose title covers at least 0.15 more of the agenda heading than the next is taken, and otherwise the roll call is marked `ambiguous` and the tied files are listed. A roll call for which no candidate reaches these thresholds, or held on a day without a matching file—a vote on an article or an amendment days before the final vote, for instance—is matched against the files whose vote by the Chamber falls in the year of the roll call or in the next three years, and accepted when the best title has at least half of its words, and at least two, in the text around the vote and is at least 0.15 ahead of the next (`title`). The object of a roll call (`kind`) is read from the words that announce it: *ensemble* or *geheel* (the whole), *amendement*, *article* or *artikel*, *ordre du jour*, *motion* or *motie*, and *urgence* or *urgentie*. A roll call on the whole of a file on the day of its vote—from 1999, a numbered vote on the whole of a bill—is flagged as `final`.

Register of members

The members of the Chamber and their mandates are reconstructed from the Dutch-language Wikipedia, which contains a list of members for every legislature since 1831. Pages are retrieved through the MediaWiki API[36], and the revision identifier of every page is stored, so that the exact version used can be retrieved. For each legislature, we parse the tables of members—name, party, district and notes—and assign each person a stable key derived from the title of the linked biography, so that the same person is recognized across legislatures. The notes record changes during a legislature: deaths, resignations, appointments and the members who replaced them. We parse them into structured fields (who replaced whom, when and why) and add, as separate rows flagged as such, the people who are named only in the notes of another member, such as members who took their seat as replacements or elected candidates who opted for another seat. Several checks guard against parsing errors: links to disambiguation pages are detected; rows that name an institution rather than a person are excluded; the sequence of legislatures is checked for gaps; and linked pages that lack the category of Belgian members of the Chamber are listed for inspection. For every member with a biography, the Dutch article and, where one exists, the French article are downloaded; download failures are recorded in the register instead of halting the run. The register covers 56 legislatures, from 1831–1833 to 2024–2029, with 10,309 member–legislature mandates and 3,138 persons, 3,133 of whom have a Dutch Wikipedia page; 3,145 Dutch and 1,979 French articles were retrieved. Because the reports print speaker labels without a title from 2000, the gender of the members is taken from Wikidata (property P21, *sex or gender*), through the item linked to each Dutch article; trans women and men are coded by their gender, and until 1999 the coding can be checked against the female titles printed in the labels. Members without a value whose first legislature began before 1929, when the first woman was elected to the Chamber, are coded as men, and the few remaining members are coded by hand from the reports or the Chamber's website (`gender_manual.csv`). Wikidata also checks the links of the register: an item that is not an instance of *human* (property P31) reveals a link to the wrong page, such as a page on a family name. Such links, and names printed without a link in the list of one legislature but linked in others, are corrected by hand in `register_fixes.csv`, which is released with the data.

Register of ministers

Speakers who address the Chamber as ministers are identified through a register of the members of the federal governments, built from the Dutch-language Wikipedia pages on the composition of every government. Each row records a ministerial term: the person, the portfolio as named on the page, the office (minister or secretary of state), the party and the dates of appointment and departure. Persons share the key of the member register when they have the same Wikipedia page. The register contains 1,751 terms held by 621 persons.

Biographical variables

Biographical variables are extracted from the Wikipedia articles by an LLM. The instructions, the schema and the article texts were first used with an open-weight model run locally through Ollama[37]: Gemma 3 with 12 billion parameters[19] (Ollama tag `gemma3:12b`, model ID `f4031aab637d`, context window of 16,384 tokens, temperature 0), which keeps the extraction free of charge and independent of commercial providers but processes only a few biographies an hour on a laptop. The released variables were extracted with the same instructions, schema and texts by Claude Haiku 4.5 (model `claude-haiku-4-5-20251001`), through the Anthropic Message Batches API, at temperature 0, with at most 8,192 output tokens and with the answer constrained by a tool schema; for the 8 members for whom it returned no valid answer, the answers of Gemma 3 are used, and the column `model` of the biographical tables records which model produced each row. The model receives the Dutch and, when available, the French article on a member and returns a JSON object whose structure is fixed by a schema (see Data Records); decoding is constrained to that schema, so every answer can be parsed. The instructions require the model to use only the text of the article; to copy names, places, degrees and occupations as written; to leave a field empty when the article is silent; and not to infer a degree from an occupation—a lawyer's law degree, for example, is recorded only if the article mentions it. Education is coded on a six-level scale, with explicit rules for the degrees of the nineteenth-century Belgian university system, which were ordinary degrees despite titles such as *doctor in de rechten*. Occupations are coded into 19 categories, and political offices other than membership of the Chamber into seven levels. The model must also state whether the article is about the member in question, which guards against homonyms, and, when the Dutch and French articles disagree, prefer the more precise statement and explain the discrepancy in a note.

Every non-empty value must be supported by a short quotation copied verbatim from the article. After the extraction, each quotation is searched for in the article and marked as found exactly, found partially or not found, so that users can discard values whose evidence cannot be located. For each member, the complete answer is stored with the digest of the model, a hash of the prompt and schema, and the revision identifiers of the articles; a member is processed again only if one of these changes. The nested answers are flattened into one table with one row per member and three long tables with one row per degree, occupation and other office. Variables are available for 3,127 persons. LLMs are increasingly used for coding tasks of this kind[38]; the evidence requirement makes their output verifiable item by item.

Record linkage

Speakers and voters are linked to the registers, following standard record-linkage practice[39]. Linkage is date-aware: for a speech or a vote on a given day, the candidates are the members of the legislature in office on that day, with missing end dates derived from the start of the next legislature and partial dates handled explicitly. The printed surname is compared with the candidates' surnames after removing case, accents and particles, so that *de Broqueville* matches *Broqueville, Charles de*. If no surname matches exactly, a surname that differs by one character (two for surnames longer than seven characters) is accepted, provided that no other candidate is equally close. A surname of several words is also found by its first word (*Philippart* for Philippart de Foy) and a double name by any of its parts (*Sneppe* for Spinnewyn-Sneppe); when several members fit, the one whose surname matches together with its particles is preferred (*Van*

Eynde is not Van den Eynde); the last part of a compound surname also identifies it (*de Montblanc* for Descantons de Montblanc); and common OCR confusions are undone (*Ilymans* for Hymans). When two members are equally close to a misread name, the one in office on the date of the sitting is chosen, and a printed initial that fits none of the candidates prevents a link. Members named only in the notes of another member take their surname from their other rows, and a given name abbreviated in the label selects among namesakes (*Jos.* and *Jul. Warnant*). The title printed in the label (*M.*, *De heer*; *Mme*, *Mevrouw*) keeps only the candidates of that gender. A printed initial, as in *A. Delporte*, selects among members who share a surname, and members who had not yet taken their seat or had already left it on the date of the sitting are discarded when another candidate remains. Candidates who declined their seat are excluded.

Speakers who are not members on that day are then sought among the ministers in office: by name, when the label names them, and by portfolio when the label gives only the office, as in *le ministre de l'Intérieur*; in the second case the portfolio printed in the label, even when misread by the OCR (*intéricur*), is compared with the portfolios of the government in office. Turns of the chair are linked through the name of the presiding officer (see Segmentation). Each row records how it was linked: `exact` or `fuzzy` for members; `minister` or `minister_role` for ministers linked by name or by portfolio, with the portfolios held on that day (`min_portfolio`); `fix` for manual decisions; and `ambiguous`, `gender_mismatch` (the printed title contradicts the gender of every candidate), `not_found` or `no_legislature` for rows that remain unlinked, the last for sittings that fall outside any legislature of the register, such as those of the National Congress. Turns of the chair whose presiding officer could not be identified are marked `chair_unnamed` and are not searched in the register; the Presidents of the Chamber are listed in `legislatures.csv`. From 2019, when a report names no presiding officer, such a turn is attributed to the President of the Chamber in office that day, provided that only one President fits the year and that the title printed in the label agrees with the President's gender (*La présidente* and *De voorzitster* for a woman, *Le président* for a man); these turns are marked `chair_president`. The rule is not applied in 2020, when the presidency changed during the year, since `legislatures.csv` records years only. Names that cannot be linked are written to a report (`unmatched_speakers.csv`), sorted by frequency and listing the three closest members of the legislature; the last run left 9,039 distinct names in it. The most frequent of them were checked by hand, and the decisions are kept in `speaker_fixes.csv`: OCR misreadings (*Heÿeis* for Hoÿois), married names and members missing from the register in one legislature. Decisions entered by hand in this report are applied in subsequent runs, and the report is released with the data, so that every manual intervention is documented. Linked rows carry the member's mandate on the date of the sitting, register entry and biographical variables.

Software

Retrieval of the debates, segmentation, diagnostics and linkage use only the Python standard library and run under Python 3.9; the conversion of the PDF reports uses PyMuPDF or pdfplumber, and the Wikipedia scrapers and the biographical extraction also use pandas, Pydantic and the Ollama client and the Anthropic Python SDK. The legislative databases are read with Selenium (Genesis) and with Playwright, Requests and Beautiful Soup (FLWB); the download of the full sheets of Genesis and the merge of the sources use only the standard library. A shell script (`run_annales.sh`) runs segmentation and linkage in sequence, and switches allow each step to be rerun on its own or the linked files to be written without the text. The sample of sittings for the gold standard is drawn with a fixed random seed (2026). The tables and figures of this article are produced by Stata do-files (Stata 16 or later) from the released files, after a helper script has counted the words of every block; `07_finnish.do` also writes the dataset on speaker characteristics, `members_year.csv`.

S2. Files and variables

Table S2 lists the files of the dataset. The dataset on speeches, the dataset on speaker characteristics and the files on legislation are described in the main text (Tables 1, 2 and 3); Tables S3–S6 describe the other files.

The linked file of votes, `votes_linked.csv`, adds to `votes.csv` the same columns as the linked file of speeches.

S3. Summary statistics

Table S7 summarizes the contents of the dataset, and Table S8 the distribution of the main quantities at the level of the speech, the sitting, the member in a legislature, the year and the member. Tables S9 and S10 break the speeches and the members down by the institutional periods of Table S1: census suffrage (to 1893), plural voting (1894–1918), equal male suffrage (1919–1944), the post-war unitary state (1945–1970), the state reforms (1971–1994) and the federal Chamber of 150 members (from 1995). The speech-level statistics use blocks of type `SPEECH` in plenary sittings; speakers are counted as distinct linked members. The dataset contains 3,403,299 blocks of debate, of which 2,365,839 are speeches (Table S7). The typical speech is short—the median speech has 29 words, the mean 167 (Table S8)—because the reports record every exchange on the floor, and a member speaks a median of 50 times in a legislature. Across periods (Table S9), the number of speeches per sitting more than triples after the introduction of plural voting and falls back after 1945; the share of speeches in Dutch rises from none in the nineteenth century to 12 % in 1919–1944 and to a majority from 1971; and the share of speeches by ministers falls from 28 % in 1830–1893 to 16 % after 1995. Among the members (Table S10), the average age at entry hardly changes, at 42 to 43 years, while the shares of nobles, lawyers and members from industry, trade and banking decline, and the share of workers and union officials rises from 1 % in the first period to 29 % in 1919–1944. Table S11 breaks the legislative files of the Chamber down by the same periods. Members' bills, rare in the nineteenth century, are about as numerous as government bills in 1945–1970 and outnumber them by about two to one in 1971–1994, while the share of budgets falls after the First World War.

S4. Further validation

Segmentation accuracy

We drew one sitting at random in each five-year period (seed 2026) and annotated by hand an excerpt of about 150 lines from each, starting at a boundary between turns. One sitting drawn for 1835–1839 turned out to be a sitting of the Senate and was left out, as it is from the data, which leaves 39 excerpts. Two of them, from 2023 and 2025, are left out of the scores until they are annotated again on the text of the PDF reports. Each block opens with an inline marker that gives its type and, for speeches, the speaker, so that annotation requires nothing more than a text editor. The excerpts were pre-annotated with the blocks found by the rules and corrected by hand by the author, following written guidelines released with the data (`GOLD_INSTRUCTIONS.md`). We score the rules with four metrics: precision, recall and F1 of the boundaries between blocks, where a predicted boundary is correct if it lies within 15 characters of a gold boundary; type accuracy, the share of matched boundaries with the right type; speaker accuracy, the share of matched speeches attributed to the right speaker; and the share of characters, page furniture excluded, that carry the right type and, inside speeches, the right speaker, which weighs errors on long speeches more heavily than errors on short interjections.

Overall, precision is 93.6 %, recall 84.7 % and F1 88.9 %; when a boundary is found, its type is right in 98.7 % of cases and its speaker in 94.8 %, and 83.4 % of the characters are attributed correctly (Table S12). A first evaluation showed three kinds of errors, which we corrected in the rules: speaker labels whose title the OCR had damaged, blocks that the OCR had run into the same line, and running heads misread by the OCR, after which the speech that continued on the next page was attributed to the clerks. We corrected them as general patterns and tested the corrections on sittings outside the sample, since corrections tailored to the excerpts would inflate the scores. The remaining errors are concentrated in the sittings of 1830–1875, the oldest and most damaged part of the source, and in boundaries missed in the reports since 1990.

To measure the reliability of the annotation, and to check that the pre-annotation does not favour the rules, a second annotator annotated 9 of the excerpts again from the raw text, without seeing the first

Table S2. Files of the dataset.

File	Unit of observation	Rows	Content
Debates			
`txt/DD_MM_YYYY.txt`	sitting	17,387	Text as retrieved
`sittings.csv`	sitting	17,280	Dates (file name, printed heading, date used), document codes, number of speeches
`skipped_files.csv`	file	107	Files that are not sittings, with the reason
`senate_files.csv`	file	–	Files of 1831–1900 by house: Chamber, Senate or both
`committee_scan.csv`	file	2,718	Files of 1985–1999: plenary sitting, committee report, Central Commission, or both a plenary sitting and a committee report
`speeches.csv`	block of the debate	3,403,299	Speeches, interjections, procedural text, documents, roll calls, summaries
`apparatus.csv`	element of the apparatus	1,077,739	Headings, presiding officers, tables of contents, page furniture, roll-call checks
Votes			
`votes.csv`	member × roll call	4,591,143	Individual votes read from the text
`vote_items.csv`	numbered vote	13,629	Dossier, document, object, outcome and counts of the votes from 1999
`votes_items.csv`	member × roll call	–	`votes.csv` with the dossier of each vote, from 1999
Legislation			
`laws.csv`	legislative file	47,924	Bills and other files of the Chamber's legislative databases: title, type, authors, votes, keywords, law
`law_documents.csv`	parliamentary document	81,051	Documents of each Genesis file
`rollcall_laws.csv`	roll call; numbered vote item	42,635	File decided by each roll call, and how it was found
Members and ministers			
`legislatures.csv`	legislature	56	Years, number of seats, source revision
`mandates.csv`	member × legislature	10,309	Party, district, roles and replacements
`politicians.csv`	person	3,138	Register of members and of their Wikipedia pages
`gender.csv`	person	3,138	Sex or gender from Wikidata
`register_fixes.csv`	correction	–	Manual corrections of the register
`ministers.csv`	ministerial term	1,751	Portfolio, office, party and dates
`bio_variables.csv`	person	3,127	Biographical variables
`education_long.csv`	degree or school	–	One row per item, with evidence
`occupations_long.csv`	occupation	–	One row per item, with evidence
`offices_long.csv`	other office	–	One row per item, with evidence
Linked files			
`speeches_linked.csv`	block of the debate	3,403,299	`speeches.csv` with member or minister, mandate and biography
`votes_linked.csv`	member × roll call	4,643,277	`votes.csv`, with the votes of 2019–2026 taken from the reports, member, mandate and biography
`members_year.csv`	member × legislature × year	46,262	Characteristics of the members, year by year (Table 2)
Documentation of quality and provenance			
`quality_by_year.csv`	year	–	Segmentation and linkage diagnostics
`link_by_decade.csv`	decade	20	Linkage rates
`speaker_fixes.csv`	name	–	Manual decisions on unlinked names
`unmatched_speakers.csv`	unlinked name	9,039	Unlinked names, closest candidates, manual decisions
`gold/DD_MM_YYYY.gold.txt`	sitting	39	Gold-standard segmentation
`extractions.jsonl`	person	3,127	Complete model answers with provenance
`topics.csv`	speech	–	Topic label and probability of every speech (in preparation)

Table S3. Variables of the apparatus of the printed record (`apparatus.csv`; one row per element).

Variable	Description
`sitting, date`	As in `speeches_linked.csv`
`kind`	`SESSION_HEAD` (heading of the sitting), `PRESIDING` (presiding officer), `SOMMAIRE` and `SOMMAIRE_ITEM` (table of contents and its items), `SECTION` (agenda heading), `RUNNING_HEAD`, `FOOTER`, `STRAY` (isolated page number), `DOCS_HEAD` (heading of the documents section), `ROLLCALL` (roll-call check)
`page`	Page on which the element is printed
`text`	Text as printed; the name of the presiding officer for `PRESIDING`; for `ROLLCALL`, the announced counts, the counts of names listed and the status
`extra`	Additional information: the function of the presiding officer, the page reference of a table-of-contents item, the number of a roll call

Table S4. Variables of the files on sittings and votes.

Variable	Description
sittings.csv — one row per sitting	
`sitting`	Identifier of the sitting
`date`	Date used throughout the dataset
`date_file, date_printed`	Date in the file name; date printed in the heading of the sitting, if found
`date_note`	Outcome of the comparison (Table S14); empty when the two agree
`speeches`	Number of blocks of type `SPEECH`
`docs`	Document codes of the reports in the file, from 1999
votes.csv — one row per member and roll call	
`sitting, date, doc`	As in `speeches.csv`
`rollcall`	Number of the roll call within the sitting, in order of appearance
`vote_no`	Official number of the vote, from 1999
`vote`	`oui` (yes), `non` (no), `abstention` or `présent`
`name`	Name normalized against the roster
`rollcall_status`	`ok`, `mismatch` or `no_counts`: agreement between the announced counts and the names listed
vote_items.csv — one row per numbered vote, from 1999	
`sitting, date, doc`	As in `speeches.csv`
`legislature`	Legislature, with `dossier` the key of a legislative file
`vote_no, item`	Official number of the vote; item, when a vote covers several texts
`dossier, doc_ref`	Number of the legislative file, and of the document within it (e.g. `3441/10`)
`doc_refs`	All the document references cited for the vote
`kind`	Object of the vote: `amendment`, `article`, `whole`, `bill`, `budget`, `motion`, `resolution`, `agenda`, `urgency`, `consideration`, `constitution`, `other`, or `?` when the title is not classified
`outcome`	`adopted` or `rejected`
`yes, no, abstentions, total`	Announced counts
`source`	Where the reference is printed: `heading`, `conclusion`, `both` or `none`
`agree`	Heading and conclusion cite the same dossier
`title, conclusion`	Heading and conclusion of the vote, as printed
votes_items.csv — `votes.csv` plus `dossier`, `doc_ref`, `kind`, `outcome`, `title` and `other_dossiers`	

Table S5. Variables of the registers of members and ministers (selection).

Variable	Description
legislatures.csv — one row per legislature	
`legislature, leg_label`	Identifier and label of the legislature (e.g. 1892–1894)
`start_year, end_year`	First and last year
`n_seats`	Number of seats
`n_rows_table, n_rows_notes`	Members parsed from the tables and from the notes
`revid`	Revision identifier of the Wikipedia page
mandates.csv — one row per member and legislature	
`person_key`	Identifier of the person (joins `politicians.csv`)
`name, surname, given_names`	Name as listed; parsed surname and given names
`legislature, leg_label`	Legislature
`party, district`	Party and electoral district, as listed
`roles`	Functions held during the legislature
`notes`	Notes as printed on the page
`row_type`	`table` (listed member) or `notes` (named only in the notes of another member)
`is_replacement, replacement_date`	Entered the Chamber as a replacement, and when
`replaces_name, replaces_reason`	Member replaced, and why that member left
`replaced_by_name`	Member who replaced this member
`left_reason, left_date`	Reason and date of departure
`has_page, link_title`	Whether the member has a Wikipedia page, and its title
politicians.csv — one row per person	
`person_key, name`	Identifier and name of the person
`has_page`	The person has a Dutch Wikipedia page
`has_mp_category`	The page carries the category of Belgian members of the Chamber
`fetch_error, fr_fetch_error`	Errors, if any, when downloading the Dutch or the French article
gender.csv — one row per person	
`person_key, wikidata_qid`	Person and Wikidata item
`human`	The item is an instance of *human* (property P31)
`gender`	`female` or `male` (Wikidata property P21; trans women and men by their gender), `other`, or empty
`gender_source`	`wikidata`, `before 1929` (first legislature before the first woman was elected) or `manual`
ministers.csv — one row per ministerial term	
`person_key`	Identifier of the person
`portfolio, office`	Portfolio as named on the page; minister or secretary of state
`party`	Party
`start_date, end_date`	Dates of appointment and departure

Table S6. Schema of the biographical variables. Every value is supported by a verbatim quotation from the article; the long tables report for each item the quotation (`evidence`) and the result of the automatic check (`evidence_check`: `exact`, `partial` or `not_found`).

Variable	Description
bio_variables.csv — one row per person	
`article_is_about_this_politician`	`yes`, `no` or `unclear`; guards against homonyms
`birth_date, birth_year, birth_place`	As written; dates in ISO format when complete
`death_date, death_year, death_place`	As written; dates in ISO format when complete
`noble_title`	E.g. *baron*, *graaf*, *burggraaf*, *ridder*, *jonkheer*
`father_name, father_occupation`	As written
`party_raw`	Party or political current, as written
`highest_education`	`none_or_primary`, `secondary`, `higher_non_university`, `university`, `postgraduate` or `unknown`
`uncertainty_note`	Ambiguities or contradictions noted by the model
education_long.csv — one row per degree or school	
`level, degree_raw, field, institution, year`	Level on the scale above; degree, discipline and institution as written; year if stated
occupations_long.csv — one row per occupation	
`title_raw, category`	Occupation as written, and one of 19 categories: lawyer; notary; magistrate; physician or pharmacist; engineer or architect; industrialist or manufacturer; merchant or trader; banker or financier; landowner or rentier; farmer; professor or teacher; journalist, publisher or writer; civil servant or diplomat; military officer; clergy; union, party or cooperative official; worker or craftsman; employee or clerk; other
`firm_or_employer, start_year, end_year`	As written; years if stated
offices_long.csv — one row per office other than membership of the Chamber	
`office_raw, level, place`	Office as written; level (municipal, provincial, senate, government, judicial, international or other); place
`start_year, end_year`	Years if stated

Table S7. Contents of the dataset.

Component	Records
Debates	
Files retrieved	17,387
left out: index volumes	80
left out: tables of contents only	27
Sittings	17,280
of which plenary	15,661
Blocks of the debate	3,403,299
speeches (`SPEECH`)	2,365,839
collective interjections (`CROWD`)	26,093
procedural text (`PROC`)	645,912
roll calls (`VOTE`)	37,626
printed documents (`DOC`)	277,357
summaries in the other language (`SUMMARY`)	47,956
Speeches linked to a member or minister (%)	95.6
Distinct speakers linked	3,073
Roll-call votes	
Roll calls detected	37,629
Roll calls with a list of names	31,487
Individual votes	4,643,277
Individual votes linked to a member (%)	95.9
Numbered votes (from 1999)	13,629
Numbered votes linked to a dossier (%)	88.6
Individual votes since 1999 linked to a dossier (%)	87.4
Members and ministers	
Legislatures	56
Member–legislature mandates	10,309
Persons in the register	3,138
with a Dutch Wikipedia article	3,133
with biographical variables	3,127
Ministerial terms	1,751
persons	621

Table S8. Distribution of the main quantities. Speeches: blocks of type `SPEECH` in plenary sittings; words are tokens separated by white space, and words and characters leave out the turns of the chair. Member and legislature: every mandate in `mandates.csv`, with zero for members who never spoke or voted. Age at the first legislature: members whose article is about them and whose age is between 18 and 90.

Variable	N	Mean	SD	P10	Median	P90
Speech						
Words per speech	1,821,629	166.7	393.9	4	29	435
Characters per speech	1,821,629	1,025.8	2,447.3	22	174	2,672
Sitting						
Speeches per plenary sitting	15,661	144.3	127.5	22	111	308
Member and legislature						
Speeches	10,306	200.3	627.7	1	50	447
Individual votes	10,306	431.8	669.3	3	109	1,420
Year						
Sittings	190	91.4	34.6	48	90	138
Roll calls	190	198.0	243.4	8	50	574
Member						
Legislatures served	3,138	3.3	2.7	1	2	7
Age at the first legislature	3,125	42.7	9.4	31	42	56

Table S9. Speeches by institutional period. Plenary sittings; speeches are blocks of type `SPEECH`. Dutch: share of speeches in Dutch among those in French or Dutch. Linked: share linked to a member or a minister. Ministers: share of speeches by a minister. Speakers: distinct members who spoke. Words, Dutch and Ministers leave out the turns of the chair.

				Words per speech					
Period	Sittings	Speeches	Per sitting	Mean	Median	Dutch (%)	Linked (%)	Ministers (%)	Speakers
1830–1893	6,496	449,213	69.2	197	36	0.0	93.5	27.5	665
1894–1918	2,385	576,704	241.8	96	17	1.4	94.2	18.1	437
1919–1944	1,895	428,738	226.2	118	23	11.7	95.5	18.9	551
1945–1970	1,954	316,192	161.8	186	27	28.7	98.7	24.1	591
1971–1994	1,412	273,976	194.0	261	46	50.8	98.7	23.2	671
1995 onwards	1,519	214,804	141.4	287	166	60.7	96.4	15.9	852

Table S10. Members by institutional period. A member is counted in every period in which he or she held a mandate; age at entry refers to members whose first legislature began in the period. Shares are computed among members whose Wikipedia article is about them; occupations whose supporting quotation was not found in the article are ignored.

				Share of members with a biography (%)					
Period	Members	With a biography	Age at entry	Noble	University degree	Lawyers	Industry, trade, bank	Land-owners	Workers, unions
1830–1893	737	735	42.5	25.0	63.0	39.5	31.0	3.5	0.7
1894–1918	422	422	42.7	20.1	63.0	38.6	21.8	4.0	11.8
1919–1944	537	537	42.7	9.5	49.9	31.3	19.4	1.3	28.7
1945–1970	578	578	42.6	5.5	49.7	24.6	13.8	0.5	27.5
1971–1994	666	660	42.4	4.8	57.9	17.9	10.3	0.0	22.3
1995 onwards	818	811	43.1	7.3	75.8	19.0	11.5	0.0	16.6

Table S11. Legislative files of the Chamber by institutional period (`laws.csv`; files of the Chamber, `K`, only: the Senate's own files are left out). A file is dated by the Chamber's vote or else by its deposit, its session, the year in its code or, failing these, the median year of the files of its legislature. Government and members' bills: types `projet de loi` and `proposition de loi`. Became law: `is_law`. Keywords: free keywords or EuroVoc descriptors.

Period	Files	Government bills (%)	Members' bills (%)	Budgets (%)	Voted by the Chamber	Became law	With keywords (%)
1830–1893	9,213	39.5	3.3	8.8	4,031	3,805	100.0
1894–1918	2,676	35.1	19.9	13.9	1,176	1,096	100.0
1919–1944	4,023	51.1	28.0	10.9	2,296	2,226	100.0
1945–1970	6,817	41.4	42.0	8.9	3,595	3,499	100.0
1971–1994	9,600	25.8	53.1	8.5	3,590	3,233	100.0
1995–2025	8,225	66.1	15.1	1.1	6,740	5,530	100.0
All	40,554	42.8	27.5	7.8	21,428	19,389	100.0

Table S12. Segmentation accuracy of the rules on the gold standard, by period. Boundaries: gold block boundaries (page furniture excluded); precision, recall and F1 of the predicted boundaries (tolerance of 15 characters); type and speaker: share of the matched boundaries with the right block type and, for speeches, the right speaker; characters: character-weighted attribution. Percentages.

			Boundaries			Accuracy		
Period	Sittings	Boundaries	Precision	Recall	F1	Type	Speaker	Characters
1830–1875	8	151	88.7	78.1	83.1	99.2	86.5	63.4
1876–1918	8	336	91.7	92.3	92.0	99.4	90.7	84.9
1919–1959	9	240	98.2	92.5	95.3	98.6	96.5	92.1
1960–1989	6	245	94.3	80.8	87.0	100.0	99.3	88.4
1990–2025	6	178	94.0	70.8	80.8	94.4	100.0	85.0
All	37	1,150	93.6	84.7	88.9	98.7	94.8	83.4

annotation or the blocks found by the rules. The two annotations agree on the boundaries with an F1 of 90.9 %, on the type of 95.1 % and on the speaker of 97.1 % of the matched boundaries, and on 82.7 % of the characters. The pre-annotation does not favour the rules: against the second annotation, made without it, the rules reach an F1 of 91.0 % and attribute 81.4 % of the characters correctly, against 88.2 % and 80.0 % against the first annotation of the same excerpts.

Linkage to the registers

Table S13 reports, by decade, how speakers and voters were linked. Overall, 95.6 % of the speeches and 95.9 % of the individual votes are linked to a member or a minister. One group of rows cannot be linked by design: the sittings of the National Congress (1830–31) precede the first legislature of the register, which explains most of the low linkage rate of the 1830s. Turns of the chair whose presiding officer is not named in the report remain unlinked, except from 2019, when they are attributed to the President of the Chamber in office that day if the title printed in the label (*La présidente*, *De voorzitster*) agrees with the President's gender; this attribution is an inference, and is marked as such (`chair_president`). It is not made in 2020, when the presidency changed during the year, nor for turns under a title of the other gender, which belong to the vice-presidents. Apart from the 1830s, 90 to 99 % of the speeches of every decade are linked, and more than 95 % of the individual votes from the 1850s onwards, except in the 1990s and 2000s (92.4 and 94.6 %). The other unlinked rows are mostly names corrupted by the OCR beyond recognition and speakers identified only by a function; the 9,039 distinct names left in `unmatched_speakers.csv` are listed by frequency, so that the most frequent can be resolved by hand first. To estimate the accuracy of the links, we checked by hand a random sample of 200 linked speeches, stratified by the six institutional periods, against the source and the member's biography: 97.5 % are linked to the right person, and every speech of the sample could be judged against the source.

Table S13. Linkage of speakers and voters to the registers, by decade (per cent). Speeches: blocks of type `SPEECH`. Exact and fuzzy: members linked by name; minister: linked to the register of ministers by name or by portfolio; manual: decisions entered in `unmatched_speakers.csv`; President: turns of the chair attributed to the President of the Chamber in office, from 2019; chair not named: turns of the chair whose presiding officer was not identified. Individual votes: one row per member and roll call.

	Speeches								Individual votes	
Decade	N	Exact	Fuzzy	Minister	Manual	President	Chair not named	Not linked	N	Linked
1830s	11,356	59.9	2.0	10.2	0.0	0.0	4.4	23.6	706	21.0
1840s	45,226	63.9	2.6	23.6	0.0	0.0	4.2	5.7	1,590	93.8
1850s	53,995	70.6	2.7	20.2	0.0	0.0	2.9	3.7	4,083	96.5
1860s	62,156	80.7	3.7	8.0	0.0	0.0	4.9	2.7	4,052	96.0
1870s	71,525	85.3	4.4	3.8	0.0	0.0	5.0	1.6	16,865	98.2
1880s	133,802	88.4	4.3	2.8	0.0	0.0	3.4	1.1	24,218	95.5
1890s	222,402	86.9	5.5	2.3	0.0	0.0	4.3	1.1	38,292	98.1
1900s	271,796	85.8	7.4	1.3	0.0	0.0	4.3	1.2	27,357	98.7
1910s	175,759	85.9	3.9	4.0	0.0	0.0	4.4	1.7	20,902	98.5
1920s	200,086	87.7	2.9	3.2	0.0	0.0	5.5	0.7	25,081	97.8
1930s	197,785	92.7	1.6	3.0	0.0	0.0	1.8	0.8	80,800	98.0
1940s	71,383	89.4	1.6	7.3	0.0	0.0	1.1	0.6	84,072	97.3
1950s	143,264	91.7	0.7	6.6	0.0	0.0	0.1	0.9	298,748	97.8
1960s	99,435	93.4	0.7	4.1	0.0	0.0	1.3	0.5	284,057	98.0
1970s	110,181	93.4	0.5	5.3	0.0	0.0	0.2	0.5	498,656	96.7
1980s	140,817	96.4	0.4	1.6	0.0	0.0	0.6	0.9	533,303	96.0
1990s	94,995	92.5	0.7	2.4	0.0	0.0	2.9	1.5	988,995	92.4
2000s	122,712	88.6	0.0	6.0	0.0	0.0	3.9	1.4	597,728	94.6
2010s	96,679	90.3	0.1	6.2	0.0	0.2	2.0	1.1	660,786	97.0
2020s	40,485	94.2	0.0	5.0	0.0	0.1	0.0	0.7	452,986	98.3
All	2,365,839	88.3	2.8	4.5	0.0	0.0	3.0	1.3	4,643,277	95.9

Table S14. Date of the sittings: date in the file name compared with the date printed in the heading of the sitting.

Date in the file name and in the printed heading	Sittings	%
The same	10,403	60.2
No date printed in the heading	4,840	28.0
Printed day differs	848	4.9
Printed year differs	665	3.8
Printed date differs	432	2.5
Day and month swapped in the file name (corrected)	92	0.5
Total	17,280	100.0

Coverage and dating of the sittings

Table S14 compares, for every sitting, the date in the file name with the date printed in its heading. The two dates agree for 60 % of the sittings. For 28 % no date could be read from the heading, either because the heading was not found or because its date is damaged beyond reading; the remaining 11 % differ in the day, the year or both, most often because the heading read by the OCR is corrupted (*4912* for 1912) or belongs to a neighbouring sitting printed in the same file, and 92 files whose name swaps day and month were corrected. The number of sittings per year (Fig. S3a) shows two low stretches outside the two world wars. From 1832 to 1843 the source contains between 15 and 175 sittings a year: the *Moniteur* years are covered as far as the digitised issues go. From 1992 to 1995 the Chamber held part of its plenary business in a Central Commission (*Commission centrale*, *Centrale Commissie*), whose proceedings were published in a separate weekly series (marked CEN) next to those of the plenary sittings (PLEN) and of the committees (COM). The files of *plenum.be* contain some of its reports, which are left out of the plenary series together with the committee reports (S7); the complete CEN series is available in the digitised archive of the Chamber and could be added in a later version. The peak of 1997–1998 came from the reports of public committee meetings, which *plenum.be* serves alongside those of the plenary sittings; they are left out as well (S7). The *Moniteur* years also contain sittings of the Senate, which are left out (S1, Screening and dating).

Diagnostics on every sitting

Because the gold standard is small, we complement it with diagnostics computed on every sitting (released as `quality_by_year.csv`). When the parser misses a speaker label, the turn that the label introduces is absorbed into the preceding block: the number of speeches per sitting falls, the share of the text that falls into speeches changes, and labels picked up in corrupted form raise the share of speakers not found in the roster. Turns of the chair whose presiding officer could not be identified are counted separately. Table S15 reports these indicators by decade. These diagnostics guided the development of the period-specific grammars described in Methods: the first pass, with the rules developed on 1880–1914, missed most labels before 1876, merged many turns between the 1960s and the 1980s, and recognized none of the modern label formats after 1990. In the final run, 97 to 100 % of the sittings of every decade contain at least one speech; the number of speeches per sitting follows Fig. 1; and the share of speakers not found in the roster is below 5 % in every decade except the 1830s (14 %), when the roll calls that supply the roster are rare. Turns of the chair whose presiding officer is not named make up between 0.1 and 5.5 % of the speeches of every decade; from 2019 most of them are attributed to the President of the Chamber at the linkage stage (see Linkage to the registers).

Roll calls

For every roll call, the announced numbers of participants and of yes and no votes are compared with the numbers of names parsed from the corresponding lists. Agreement shows that the lists were split

Table S15. Segmentation diagnostics by decade, from `quality_by_year.csv`. With a speech: sittings with at least one block of type `SPEECH`. Text in speeches: share of the characters of a sitting that fall in speeches (average over years, weighted by sittings). Not in roster: speeches whose speaker is not found in the roster of the period. Chair not named: turns of the chair whose presiding officer was not identified. Counts differ: roll calls whose announced counts differ from the names listed, as read from the text; from 2019 the released votes come from the Chamber's HTML reports and from *mijnkamer.be*, whose lists agree with the counts (Table S16).

Decade	Sittings	With a speech (%)	Speeches per sitting	Text in speeches (%)	Not in roster (%)	Chair not named (%)	Roll calls	Counts differ (%)
1830s	676	97.0	17.6	61.5	13.7	4.1	7	42.9
1840s	1,005	98.4	46.2	65.9	3.3	4.5	164	9.8
1850s	1,212	98.4	44.6	53.8	3.1	2.9	226	15.5
1860s	1,099	98.5	56.6	70.0	1.6	4.9	175	15.4
1870s	1,059	98.5	67.5	77.4	2.2	5.0	299	42.5
1880s	1,065	99.7	125.6	79.5	3.2	3.4	317	70.0
1890s	1,234	100.0	180.2	81.5	2.1	4.3	434	64.7
1900s	1,118	99.9	243.1	79.2	2.8	4.3	275	70.2
1910s	575	99.8	305.7	78.2	2.5	4.5	171	78.4
1920s	972	99.9	205.8	73.5	3.3	5.5	232	63.8
1930s	792	99.5	249.7	68.2	2.4	1.8	765	52.9
1940s	453	98.9	157.6	64.9	4.8	1.1	637	67.3
1950s	791	99.9	181.1	65.2	4.5	0.1	2,415	42.8
1960s	682	99.6	145.8	62.4	3.5	1.3	3,626	32.0
1970s	688	100.0	160.1	69.3	3.5	0.2	4,599	36.7
1980s	666	99.5	211.4	70.0	1.5	0.6	4,166	30.7
1990s	801	99.9	118.6	72.0	0.1	2.9	5,941	0.0
2000s	1,173	100.0	104.6	66.3	0.0	3.9	4,564	0.4
2010s	1,032	99.9	93.7	65.1	0.0	2.3	4,976	0.1
2020s	274	100.0	147.8	74.9	0.0	0.1	3,640	0.5

correctly and that no names were lost to OCR errors; since the voter lists also supply the roster used to normalize speaker names, the check guards the speech data as well. Table S16 reports the outcome by decade. Before 2000 the announced counts and the lists agree exactly in fewer than half of the roll calls, but most discrepancies are small: in the roll calls that announce a total, the lists usually contain between 90 and 110 % of the announced names, which points to a few names lost or split by the OCR rather than to lists read wrongly (`diagnosi.txt`). In the 2000s the counts agree for 63 % of the roll calls with a list, and for 37 % the rules find no counts, although the reports print them; from the 2010s they agree for more than 99 %, and in the 2020s the lists come from the HTML reports and from *mijnkamer.be*. In the 1990s no counts are found next to the lists, which are printed in a bilingual annex, so that the roll calls of that decade cannot be checked in this way; and before the 1870s many roll calls either print no counts or have no list of names.

Register of members

The register passed three checks. The sequence of legislatures has no gaps: the apparent gap between 1985–1987 and 1988–1991 reflects the naming of the page after the election of 13 December 1987. Of the linked pages, 52 lacked the category of Belgian members of the Chamber: 46 belonged to people named only in the notes of other members, one was an institution parsed as a person, which the parser now excludes, and the remaining five were members whose pages lack the category.

Votes and legislative files

Table S17 reports, by legislature, the numbered votes, the share linked to a dossier and the number of votes whose heading and conclusion cite different dossiers, a direct check on the attribution of references to votes. Table S18 reports the objects of the votes and where their references are printed. A vote remains unlinked when the report prints no document number next to it—for instance, votes on motions—or when

Table S16. Roll calls by decade. Detected: roll calls found in the text. With a list: roll calls with at least one individual vote. The last three columns give the outcome of the comparison between the announced counts and the names listed, as a share of the roll calls with a list.

Decade	Roll calls detected	With a list of names	Roll calls with a list (%)		
			Counts agree	Counts differ	No counts printed
1830s	7	5	0.0	60.0	40.0
1840s	164	33	3.0	48.5	48.5
1850s	226	84	9.5	41.7	48.8
1860s	175	55	14.5	49.1	36.4
1870s	299	220	29.5	57.7	12.7
1880s	317	277	14.8	80.1	5.1
1890s	434	382	25.7	73.6	0.8
1900s	275	251	23.1	76.9	0.0
1910s	171	162	17.3	82.7	0.0
1920s	232	196	23.0	75.5	1.5
1930s	765	568	27.8	71.3	0.9
1940s	637	564	22.5	76.1	1.4
1950s	2,415	1,736	39.7	59.6	0.7
1960s	3,626	1,896	37.9	61.2	0.8
1970s	4,599	3,118	45.3	54.2	0.5
1980s	4,166	3,262	47.0	39.2	13.8
1990s	5,941	5,840	0.0	0.0	100.0
2000s	4,564	4,527	62.6	0.4	36.9
2010s	4,976	4,959	99.9	0.1	0.0
2020s	3,640	3,352	99.6	0.4	0.0

the number is printed in a form that the rules do not recognize. Overall, 88.6 % of the numbered votes are linked to a dossier. The share is highest in the 53rd legislature (97 %) and lowest in the 50th and 51st (1999–2007, 80 and 81 %), and heading and conclusion cite different dossiers for only 57 votes. By object, bills, articles, amendments, resolutions and votes on the whole are almost always linked, whereas motions, which concern no document, almost never are; the votes classified as `other`, more than half of the total, are linked in 86 % of cases. In a random sample of 100 linked votes, stratified by legislature and checked by hand against the pages of the dossiers on the Chamber's website, the dossier is the right one in 95.0 % of cases.

Roll calls and legislative files

Table S19 reports, by decade, the two sides of the link between roll calls and legislative files. On the side of the roll calls, 80.9 % are linked to a file: 90.3 % of the items of the numbered votes from 1999, through the number printed next to them, and 73.4 % of the roll calls before 1999, through the date and the title—11,905 by the date of the vote and 1,443 by the title alone. Before 1999, 89.7 % of the roll calls on the whole of a bill are linked, against 36.4 % of those on motions, which mostly concern no file; 544 roll calls remain tied between files voted on the same day, which are listed with them.

On the side of the files, every file voted by the Chamber should have a roll call on the day of its vote (see Institutional setting). From 1999, 99.6 % of the files voted have a roll call on that day and 98.8 % are linked to one; before 1999, 73.7 % and 43.7 %. Before 1999 the share is highest in 1930–1979, when 96.7 % of the files voted have a roll call on the day of the vote and 68.0 % are linked. Before 1930 only 43.9 % of the files voted have a roll call on that day and 18.5 % are linked: the roll calls of these decades are incompletely detected in the text, so that for them the roll calls of Table S16 and the individual votes are lower bounds. In 1980–1998, 92.2 % of the files voted have a roll call on that day, but only 48.9 % are linked.

Table S17. Numbered votes linked to a legislative file (dossier), by legislature. Heading and conclusion differ: votes whose heading and conclusion cite different dossiers.

Legislature	Years	Numbered votes	With a dossier (%)	Heading and conclusion differ
49	1999–1999	467	90.1	6
50	1999–2003	2,160	79.6	1
51	2003–2007	1,790	81.3	20
52	2007–2010	999	86.4	4
53	2010–2014	1,867	96.9	20
54	2014–2019	2,692	94.4	6
55	2019–2024	2,983	89.6	0
56	2024–2025	671	89.3	0
All	1999–2025	13,629	88.6	57

Table S18. Numbered votes by object of the vote and by the place where the document number is printed.

	Numbered votes	%	With a dossier (%)
Object of the vote (`kind`)			
`other`	7,946	58.3	86.3
`amendment`	3,130	23.0	99.7
`article`	961	7.1	99.8
`bill`	452	3.3	99.6
`motion`	401	2.9	1.2
`budget`	194	1.4	96.9
`whole`	175	1.3	99.4
`resolution`	118	0.9	99.2
`agenda`	92	0.7	98.9
`constitution`	66	0.5	100.0
`?`	60	0.4	68.3
`urgency`	20	0.1	60.0
`consideration`	14	0.1	7.1
Document number printed in (`source`)			
`heading`	5,940	43.6	100.0
`criv html (heading)`	3,027	22.2	100.0
`both`	2,863	21.0	100.0
`none`	1,189	8.7	1.5
`criv html (none)`	372	2.7	0.5
`mijnkamer`	163	1.2	94.5
`conclusion`	71	0.5	100.0
`criv html (both)`	3	0.0	100.0
`criv html (conclusion)`	1	0.0	100.0

Table S19. Roll calls and legislative files, by decade. Files voted by the Chamber: files whose sheet gives the date of the Chamber's vote, by decade of that date; roll call that day: the data contain a roll call on that date; linked: at least one roll call is linked to the file. Roll calls: before 1999, the roll calls found in the text; from 1999, the items of the numbered votes (`rollcall_laws.csv`). Linked: share linked to a file, by the date of the vote (with or without its title), by the title alone or by the document number printed next to the vote, which add up to the share linked; ambiguous: tied between files voted on the same day, and not linked. Percentages, except N.

	Files voted by the Chamber			Roll calls (from 1999: items of numbered votes)					
Decade	N	Roll call that day	Linked to a roll call	N	Linked	By vote date	By title	By document number	Ambiguous
1830s	433	0.2	0.0	7	0.0	0.0	0.0	0.0	0.0
1840s	754	30.8	12.6	161	66.5	62.7	3.7	0.0	0.0
1850s	703	46.9	22.8	226	74.3	68.1	6.2	0.0	0.0
1860s	643	48.8	18.2	175	68.0	64.6	3.4	0.0	2.3
1870s	685	64.2	25.7	299	66.6	61.5	5.0	0.0	0.0
1880s	611	66.6	33.6	317	75.7	66.6	9.1	0.0	0.0
1890s	618	67.0	36.1	434	65.7	54.8	10.8	0.0	1.4
1900s	483	38.9	19.5	275	52.4	34.5	17.8	0.0	0.0
1910s	469	37.1	11.5	171	58.5	36.3	22.2	0.0	0.6
1920s	1,108	32.0	7.4	232	50.0	35.3	14.7	0.0	9.1
1930s	1,004	93.9	46.8	765	71.2	65.1	6.1	0.0	6.1
1940s	587	97.1	69.3	637	79.3	72.2	7.1	0.0	2.7
1950s	1,459	97.4	70.7	2,415	81.0	75.7	5.3	0.0	2.9
1960s	1,428	97.7	75.2	3,626	83.7	79.0	4.7	0.0	2.0
1970s	1,644	96.9	72.0	4,599	80.5	69.2	11.3	0.0	3.3
1980s	1,398	92.0	31.3	4,166	48.1	41.8	6.2	0.0	3.7
1990s	1,781	93.4	67.9	5,990	81.3	1.5	0.6	79.2	0.0
2000s	2,031	98.9	99.2	6,691	85.6	0.0	0.0	85.6	0.0
2010s	2,299	100.0	99.7	7,082	96.5	0.0	0.0	96.5	0.0
2020s	1,366	99.9	100.0	4,367	87.4	0.0	0.0	87.4	0.0
All	21,504	80.9	59.0	42,635	80.9	27.9	3.4	49.5	1.3

Table S20. Validation of the biographical variables. (a) Share of extracted items whose supporting quotation is found in the article. (b) Coverage and plausibility of the member-level variables.

(a) Evidence check

		Quotation found in the article (%)		
Items	N	Exactly	Partially	Not found
Education	3,817	97.9	1.9	0.2
Occupations	8,919	97.9	1.7	0.4
Offices	14,921	96.6	2.7	0.7

(b) Coverage and plausibility

Members	N
With biographical variables	3,127
article about the member (`yes`)	3,125
Among the latter, share with (%)	
Year of birth	100.0
Place of birth	99.6
Year of death	72.1
Noble title	11.8
Level of education stated	74.8
Age at first legislature outside 18–90	0.0

Biographical variables

We assess the biographical variables in two ways (Table S20). First, the automatic evidence check reports, for each extracted item, whether its supporting quotation occurs in the article. Second, we check the coverage of the member-level variables and the plausibility of the birth years against the dates of the mandates: a birth year that implies an age below 18 or above 90 at the first legislature signals an error of extraction or of linkage.

The quotations that support the extracted items are found exactly in the article for 97 to 98 % of the degrees, occupations and other offices, and are not found for fewer than 1 % (Table S20a). Of the 3,127 members with biographical variables, 99.9 % have an article about them rather than about a namesake; among them, year and place of birth are known for almost all, the year of death for 72 %, a noble title for 12 % and the level of education for 75 %, and no birth year implies an implausible age at the first legislature (Table S20b).

Language of the blocks

The language assigned to each block can be checked against the language of the speaker's party after the split of the national parties along linguistic lines. From 1975, members of Dutch-speaking parties speak Dutch in 87 to 97 % of their speeches, and members of French-speaking parties in 0 to 10 % (`06_checks.do`). Members also speak the other national language at times, so these shares bound the error of the rule from above rather than measure it.

S5. Stylized facts

The number and length of the speeches and the characteristics of the members are shown in Figs. 1 and 2 of the main text; Figs. S3–S7 describe the data along other dimensions most used in research on parliaments. Because several features of Belgian parliamentary history are known in advance, they also serve as face-validity checks. Each figure marks the reforms of 1894, 1900, 1919, 1949 and 1995 with dashed lines and

covers the complete years up to 2025; annual series are interrupted, not bridged, in years without sittings. Shares and lengths of speeches leave out the turns of the chair: attributed to the presiding officer, they are so numerous that otherwise the party of the President of the Chamber would dominate the speeches of members in several legislatures, and the gender of the President the share of speeches by women.

Activity. The Chamber should have almost stopped sitting during the German occupations of 1914–1918 and 1940–1944, and its activity should respond to elections and dissolutions (Fig. S3). The number of roll calls reflects both legislative output—every bill was voted on as a whole by roll call (see Institutional setting)—and the roll calls taken on articles, amendments and motions. The number of sittings (Panel a) is stable at 100 to 175 a year from the mid-1840s to 1914, falls to between 50 and 90 a year from the Second World War to the mid-1980s and to between 40 and 70 in most years afterwards, once the reports of committee meetings are left out, and drops to zero in the war years. Roll calls detected in the text (Panel c) number a few dozen a year until the 1920s, fewer than the bills voted in the same years, which shows that the roll calls of this period are incompletely detected (S4, Roll calls and legislative files); they become frequent after the Second World War, with 600 to 800 a year in the busiest years from 1970 on, almost 1,200 in 1993 and more than 900 in 1998.

Language and gender. The share of speeches in Dutch should be negligible in the nineteenth century and grow after the Equality Law of 1898 and the language legislation of the 1930s (Fig. S4a). Women should appear among the speakers only after the election of the first woman member in 1929 (Fig. S4b). Because the labels print no title from 2000, the figure uses the gender recorded in Wikidata for the linked speakers, and compares the share of speeches by women with the share of women among the members of each legislature. Speeches in Dutch are absent before the mid-1890s, reach a few per cent before 1914, rise to about a quarter of the speeches by the late 1930s and again after the Second World War, and form the majority from the 1980s, with a peak of about 70 % around 2000. Speeches by women are rare until the 1970s; from then on they follow the share of women among the members closely, though below it, as that share rises after 1995 to about 40 %.

Parties. The party composition of speech should mirror the known balance of forces (Fig. S5): the Catholic majority that governed alone from 1884 to 1914, the entry of the Belgian Workers' Party in 1894, the first Flemish nationalists in 1919, and the split of the national parties along linguistic lines from the late 1960s, after which the families group their Dutch- and French-speaking successors. Until 1894 Catholics and Liberals share the speeches of members between them, at 40 to 60 % each. In 1894 the Liberals, reduced to a small group by the first elections under plural voting, fall below 10 %, the Socialists enter with about a third of the speeches and the Catholic share peaks at over 60 %; proportional representation restores the Liberals to about a fifth from 1900. In the interwar years the Socialists and the Catholics are the largest families, with 30 to 45 % of the speeches each, and Flemish nationalists appear from 1919; from the 1970s the other families and the labels that are not recognized grow to about a quarter of the speeches.

Party families. Party labels change over time and are recorded as listed. The party families used in Fig. S5 and in `members_year.csv` are a coarse harmonization for descriptive purposes, released with the do-files; users who need a finer classification should build their own from `party` in `mandates.csv`.

Who speaks. Fig. S6 reports, for each legislature, the share of its members who spoke at least once and the share of speeches delivered by the most active tenth of the speakers, a simple measure of how concentrated the floor is. The turns of the chair are left out of the second measure, because the presiding officer intervenes far more often than any other member and is identified more completely in some periods than in others. From the mid-1840s, 90 to 98 % of the members of most legislatures speak at least once; the lower values of the 1830s reflect in part the lower linkage rate of that decade, and the dips of single legislatures, such as that of 1949–1950, their short duration. The floor was most concentrated in the second

half of the nineteenth century, when the most active tenth of the speakers delivered 50 to 62 % of the speeches, and is least concentrated after 1990, at 30 to 47 %.

Members. Finally, the social profile of the members should shift after the extensions of the suffrage (Fig. S7): fewer nobles and fewer members with a university degree, fewer lawyers, and more workers and union officials after 1894 and 1919. Because the biographies are more often missing for some members than for others, the figure shows only legislatures in which at least 80 % of the members have biographical variables. The expected shifts are visible: the shares of nobles and of members from industry, trade and banking fall over the whole period; lawyers fall from 40 to 50 % before 1914 to below 20 % today; and workers and union officials rise from almost none to about 13 % after 1894 and 30 to 35 % after 1919, before declining from the 1960s. The share with a university degree falls after 1894 and 1919 and rises again from the late 1960s to about 80 %. The mean age at the start of the legislature rises from 43 in the 1830s to above 50 in the late nineteenth century, drops with the enlarged electorate of 1894, and declines after 1970 to about 45 to 47.

S6. Modules in preparation

Senate module, 1880–1914 (in preparation)

A module for the Senate is in preparation and will be released in a later version. The two houses were elected under different rules and were exposed differently to the reforms of 1893 and 1899, so that the Senate is a natural comparison group for the Chamber[10]. The module will apply the same pipeline to the Senate sittings of 1880–1914 and link them to a register of senators; its sources, coverage and validation will be documented with its release.

Topic labels (in preparation)

This module is not yet part of the release; it will be added in a later version. We plan to label every speech with a policy topic by a classifier trained in three steps, following the approach of the companion study of 1880–1912[10]. First, an LLM proposes topics from the items of the tables of contents and the agenda headings, which describe the business of each sitting in a few words and are available throughout the period (`SOMMAIRE_ITEM` and `SECTION` rows of the apparatus table); the proposed topics are grouped by hand into a codebook of macro-topics. Second, an LLM labels a stratified sample of speeches with the codebook, and a sample of these labels is checked by human coders. Third, the labelled sample trains transformer classifiers for French and Dutch—CamemBERT[40] and RobBERT[41]—which label every speech. To support international comparisons, the macro-topics will be mapped to the major topics of the Comparative Agendas Project[42]. The codebook, the sample sizes, the treatment of the bilingual reports and the performance of the classifiers will be documented with the release of the labels, together with their validation: agreement between LLM and human labels on the checked sample, and precision, recall and F1 of the classifiers by topic, language and period on held-out data.

S7. Notes for users

Choosing the unit of analysis. Most analyses of legislative speech will use blocks of type `SPEECH` whose speaker is linked to a member (`link` equal to `exact`, `fuzzy` or `fix`) or to a minister (`minister`, `minister_role`). `CROWD` and `PROC` blocks are useful for studies of procedure and heckling but should not be attributed to individual members. `DOC` blocks contain texts read into the record, such as the articles of bills, and `SUMMARY` blocks paraphrase an intervention in the other language: both should be excluded from measures of what members said. Turns of the chair are attributed to the presiding officer; from 2019 some are attributed by inference to the President of the Chamber in office (`chair_president`), and

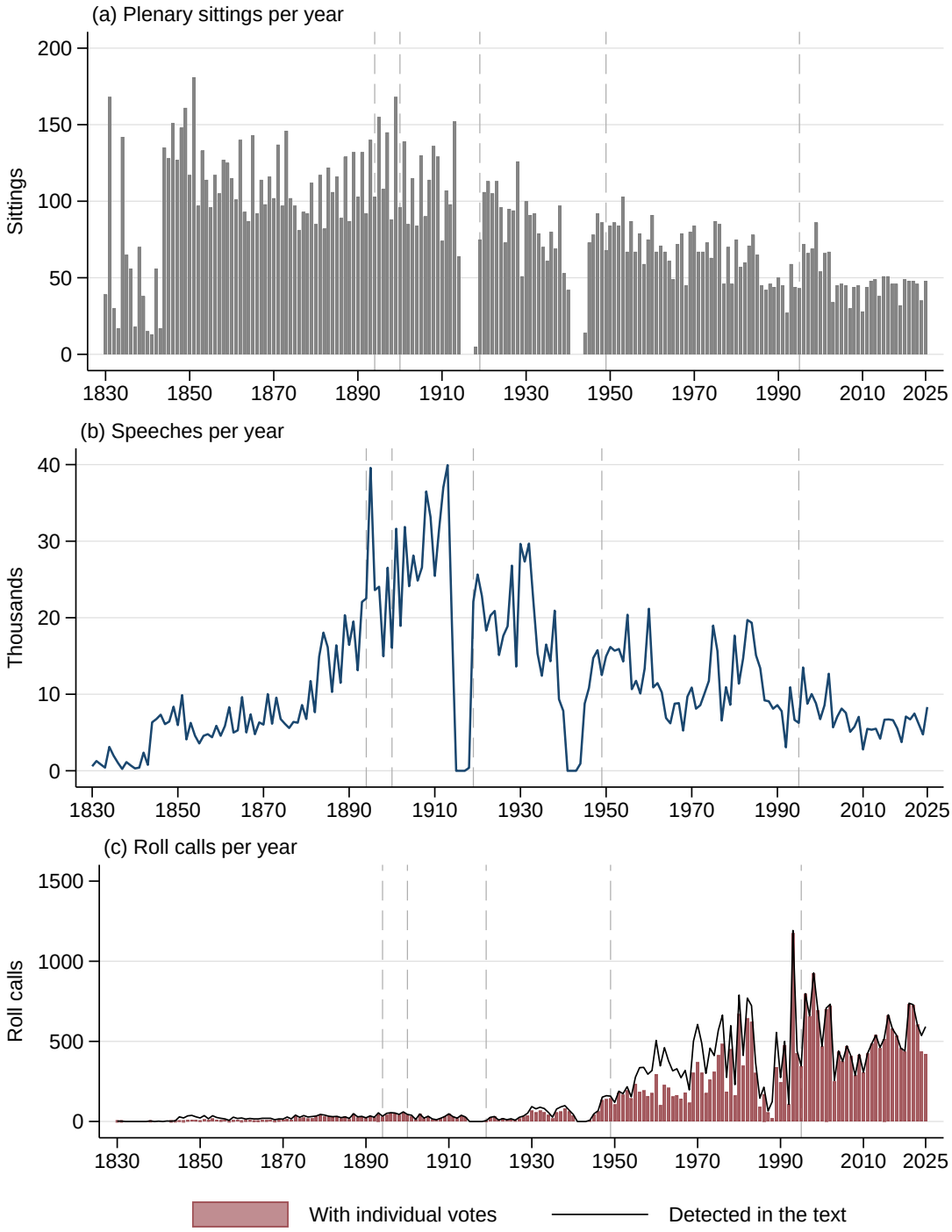


Fig. S3. Parliamentary activity, 1831–2025. (a) Plenary sittings per year. (b) Speeches per year (blocks of type `SPEECH`, thousands). (c) Roll calls per year: with individual votes (bars) and detected in the text (line). Years in which the Chamber did not sit count as zero.

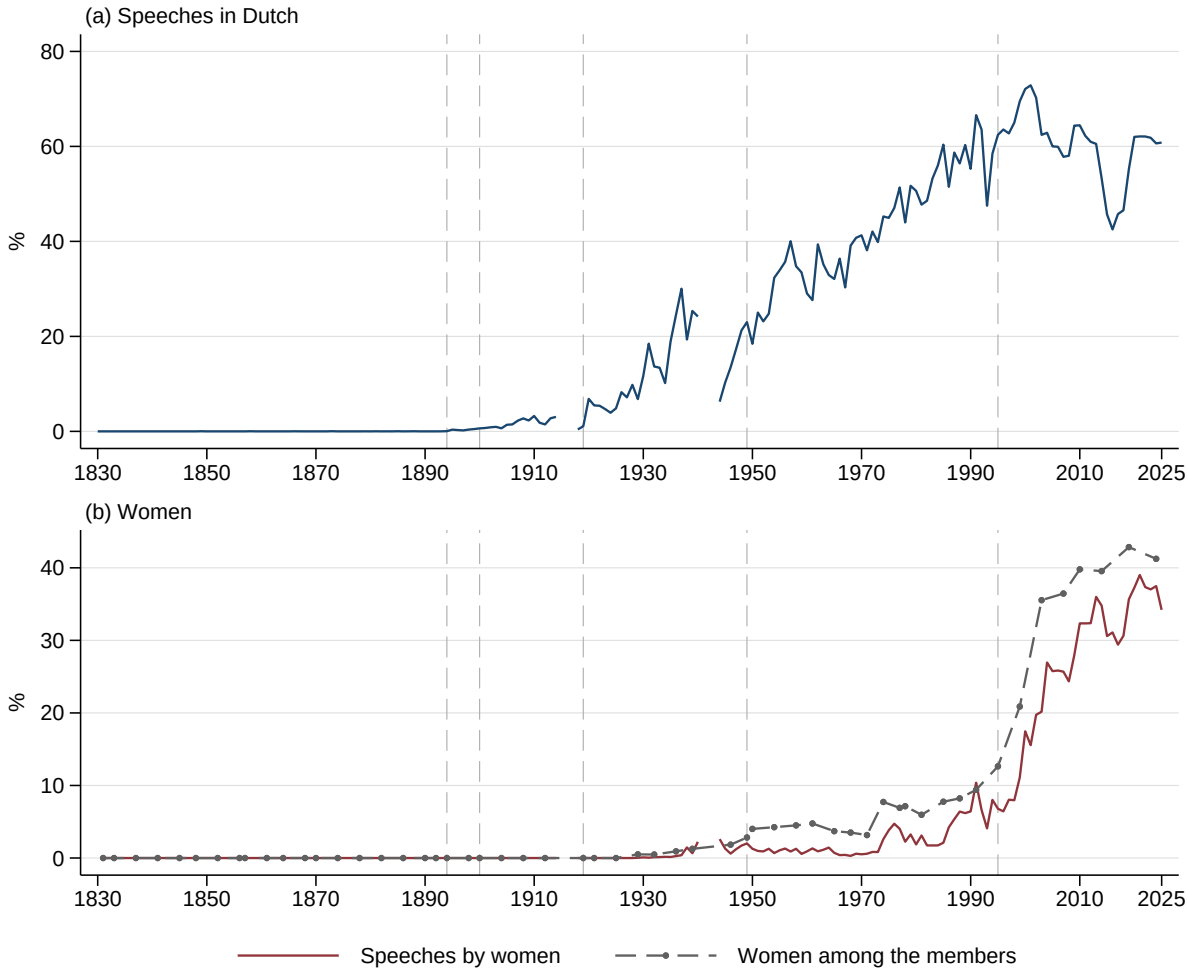


Fig. S4. Language and gender. (a) Share of speeches in Dutch among speeches in French or Dutch. (b) Share of the speeches of linked members delivered by women, by year, and share of women among the members of each legislature; gender from Wikidata. Turns of the chair are left out.

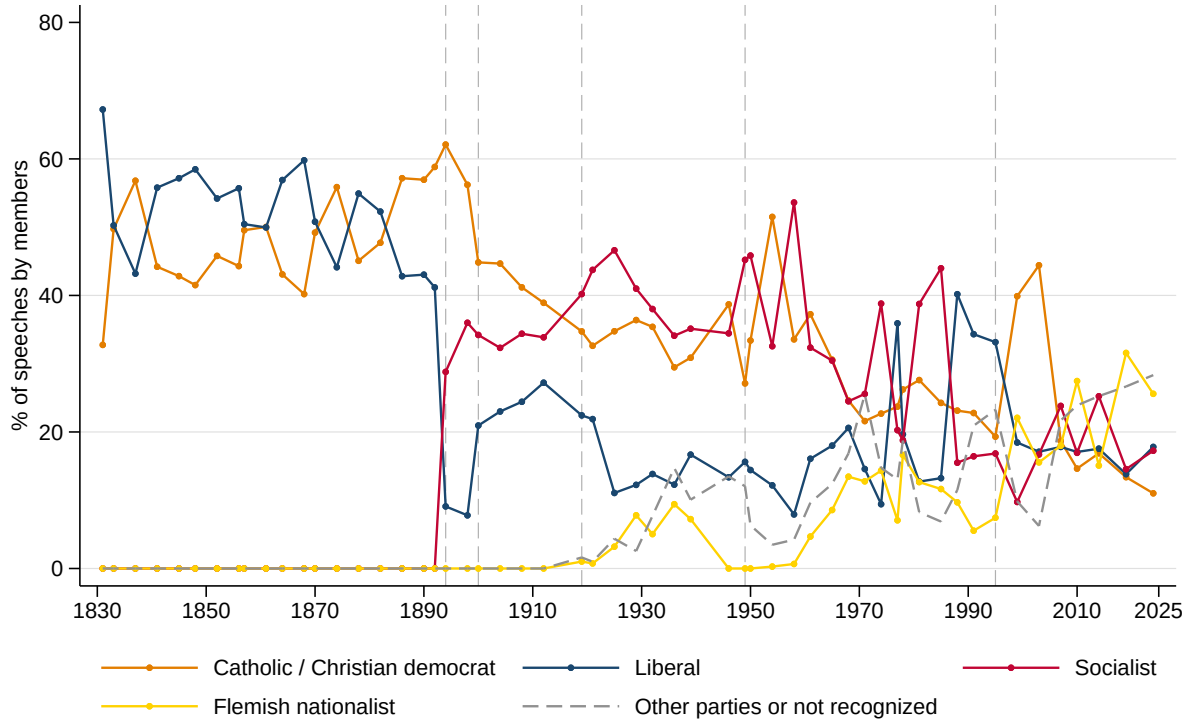


Fig. S5. Share of the speeches of members held by the four largest party families, by legislature. The party is the member's party in the register for that legislature or, for the modern period, the party printed in the label. Dashed: all other parties, and party labels that the harmonization does not recognize. Turns of the chair are left out.

users who need only names printed in the report can leave them out. Speeches can be grouped into debates through the agenda heading stored in `section`.

Coverage and quality. From November 1985 the files also contain the reports of public committee meetings; from 1999 the document code in `doc` identifies them (`PLEN` or `COM`), and before 1999 the title of each report does (see below). The tables and figures of this article use plenary sittings only. The 27 files of the 1990s that contain only a table of contents leave their sittings without debate text; the sittings of the National Congress (1830–31) are included but not linked; and the sittings of the Senate printed in the *Moniteur belge* of 1831–1848 and the Central Commission of 1992–1995 are left out, as are the few sittings that *plenum.be* reports as missing. Fourteen plenary sittings of February and March 2003, which *plenum.be* does not list, are missing, and the file of 27 January 2000 holds a sitting of 28 June 2001 instead of that of its date. The text of the blocks is not corrected: OCR errors are frequent in the older volumes and affect word counts and dictionary-based measures[43], and any passage can be checked against the facsimiles on *plenum.be* through the pages stored with each block. Votes whose roll call has `rollcall_status` equal to `mismatch` come from lists in which names were missed or misread, and analyses of individual voting should check that their results hold without them. From 1999, `vote_items.csv` links the numbered votes to their legislative files, which are identified by `legislature` and `dossier`, since numbers restart in every legislature. Differences in quality across periods are documented in Tables S12 and S13 and in Supplementary Information S4, and users should restrict or weight their samples accordingly.

Roll calls and legislative files. To study how members voted on legislation, `rollcall_laws.csv` can be joined to `votes_linked.csv` on `sitting` and `rollcall`, and to `laws.csv` on `law_id`. The method of each link is recorded: links by document number (from 1999) and by the date of the vote rest on the official record of the vote, while links by the title alone concern roll calls held on another day



Fig. S6. Who speaks, by legislature. (a) Share of the members of the legislature who spoke at least once. (b) Share of the speeches of members, turns of the chair excluded, delivered by the most active 10 % of the speakers.

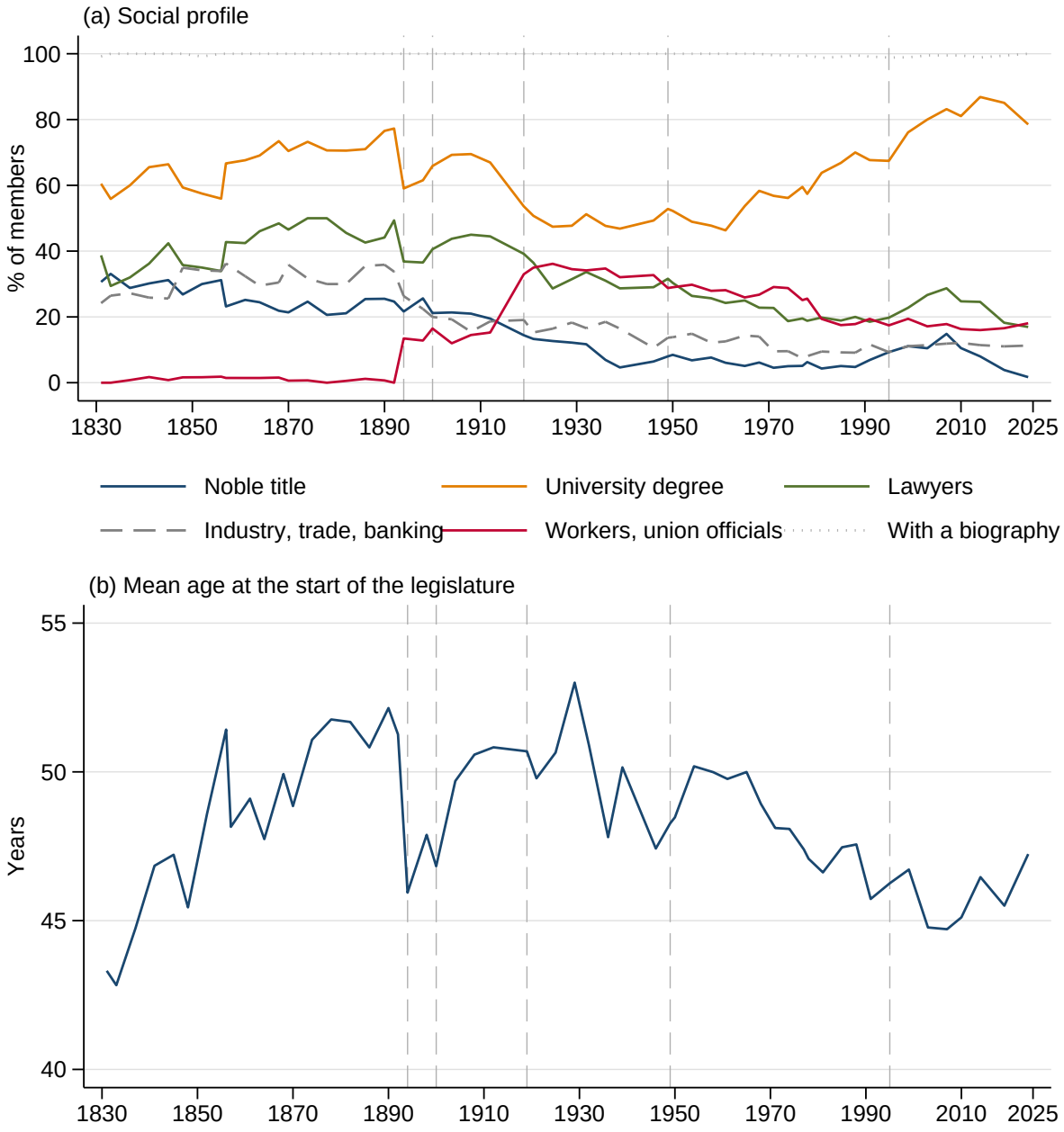


Fig. S7. Members of the Chamber, by legislature. (a) Shares with a noble title, with a university degree, and with an occupation as lawyer, in industry, trade or banking, and as a worker or union official, among members whose article is about them; dotted, the share of members with biographical variables. (b) Mean age at the start of the legislature. Legislatures in which fewer than 80 % of the members have biographical variables are not shown.

than the Chamber's vote on the file and rest on the words of the title; users can leave the latter out or require a higher `score`. Rows with `final` equal to 1 are votes on the whole of a file; some files have several, among which users who need one vote per file must choose. Before 1930 many roll calls are missing from the data (S4), so that many files voted in those years have none. The subject keywords differ between the two sources—free keywords in Genesis, mostly EuroVoc descriptors in the FLWB—so that analyses of the subjects of legislation across 1988 need a mapping between the two.

Biographical variables. The biographical variables are only as complete as the Wikipedia articles, which are more detailed for prominent members: a missing value means that the article is silent, not that the member lacked the attribute. We recommend discarding values whose evidence check is `not_found` and treating all variables as missing when `article_is_about_this_politician` is not `yes`. The column `model` records the model that produced each row (Claude Haiku 4.5 or, for 8 members, Gemma 3).

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Committee meetings before 1999. Before 1999 the reports print no document code, and *plenum.be* serves the reports of the public committee meetings and, in 1992–1995, those of the Central Commission in the same way as those of the plenary sittings. Each report opens with its title, which we read in the first lines of every file of 1985–1999: *Réunion publique de commission* or *Handelingen van de openbare commissievergaderingen*, or a committee document number (*N. C4*), for a committee meeting; *Séance plénière* or *Handelingen van de plenaire vergaderingen* for a plenary sitting; and *Centrale Commissie* or a document number *CC* for the Central Commission. When no such title is printed, the name of a committee at the head of the report (*COMMISSION DE LA JUSTICE*) or the number and the date of a plenary sitting (*N. 41*, *Séance du mardi ...*) decide, and a title always prevails over them, so that a plenary sitting whose table of contents names a committee stays plenary. Only lines in capitals or short lines count, so that a committee mentioned in a debate does not. The 300 committee reports and the 12 reports of the Central Commission found in this way are left out of the plenary series used in this article; they account for the rise in the number of files in 1997–1998. Because the files are named by date, a plenary sitting and a committee meeting held on the same day could end up in one file; we look for the title of the other kind of report later in each file, on a line of its own, and find none. No file before November 1985 is a committee report. The classification of every file is released as `committee_scan.csv`, so that users can apply it to the speeches.